

Cross-modal interaction of Alpha Activity does not reflect inhibition of early sensory processing: A frequency tagging study using EEG and MEG

Reviewed Preprint

v1 • March 7, 2025

Not revised

Marion Brickwedde , Rupali Limachya, Roksana Markiewicz, Emma Sutton, Christopher Postzich, Kimron Shapiro, Ole Jensen, Ali Mazaheri

Department of Child and Adolescent Psychiatry, Charité - Universitätsmedizin Berlin, Berlin, Germany • Physikalisch Technische Bundesanstalt, Berlin, Germany • Centre for Human Brain Health, School of Psychology, University of Birmingham, Birmingham, United Kingdom • Istituto Italiano di Tecnologia, Genova, Italy • Max-Planck Research Institute, Leipzig, Germany

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This manuscript addresses the role of alpha oscillations in sensory gain control. The authors use an attention-cuing task in an initial EEG study followed by a separate MEG replication study to demonstrate that whilst (occipital) alpha oscillations are increased when anticipating an auditory target, so is visual responsiveness as assessed with frequency tagging. The authors propose their results demonstrate a general vigilance effect on sensory processing and offer a re-interpretation of the inhibitory role of the alpha rhythm. While these results are **valuable**, the provided evidence is **incomplete**.

<https://doi.org/10.7554/eLife.106050.1.sa4>

Abstract

Selective attention involves prioritizing relevant sensory input while suppressing irrelevant stimuli. It has been proposed that oscillatory alpha-band activity (~10 Hz) aids this process by functionally inhibiting early sensory regions. However, recent studies have challenged this notion. Our EEG and MEG studies aimed to investigate whether alpha oscillations serve as a ‘gatekeeper’ for downstream signal transmission. We first observed these effects in an EEG study and then replicated them using MEG, which allowed us to localize the sources.

We employed a cross-modal paradigm where visual cues indicated whether upcoming targets required visual or auditory discrimination. To assess inhibition, we utilized frequency-tagging, simultaneously flickering the fixation cross at 36 Hz and playing amplitude-modulated white noise at 40 Hz during the cue-to-target interval.

Consistent with prior research, we observed an increase in posterior alpha activity following cues signalling auditory targets. However, remarkably, both visual and auditory frequency tagged responses amplified in anticipation of auditory targets, correlating with alpha activity

amplitude. Our findings suggest that when attention shifts to auditory processing, the visual stream remains responsive and is not hindered by occipital alpha activity. This implies that alpha modulation does not solely regulate ‘gain control’ in early sensory areas but rather orchestrates signal transmission to later stages of the processing stream.

Introduction

In our daily life, we are often confronted with sensory information from many different sources, all at once. To operate effectively, we require selective attention, reconciling the tension between environmental inputs relevant for top-down goals and sensory information that may be perceptually salient but task-irrelevant. Previous research has suggested that oscillatory activity in the alpha range (~10 Hz) plays a mechanistic role in selective attention through functional inhibition of irrelevant cortices (see **Fig. 1A**; Foxe et al., 1998; Jensen & Mazaheri, 2010; Klimesch et al., 2007). The concept of functional inhibition refers to an area of the cortex being actively hindered to process input, which is distinctly different from idling, where a part of the cortex is not actively involved. Evidence supporting the functional inhibition framework of alpha modulation revealed an increase in alpha-power over task-irrelevant sensory cortices after the onset of cues indicating the spatial location (Kelly et al., 2006; Okazaki et al., 2014; Thut et al., 2006; Worden et al., 2000; Zumer et al., 2014) or specific modality of an upcoming target (Foxe et al., 1998; Fu et al., 2001; Mazaheri et al., 2014). Moreover, previous investigations have observed ‘spontaneous fluctuations’ of alpha power in sensory regions, particularly in the visual cortex, to be inversely related to discrimination ability (Ergenoglu et al., 2004; Van Dijk et al., 2008). Alpha inhibition is believed to be transmitted in a phasic manner, as phosphene perception as well as high-frequency and spiking activity vary in line with the alpha cycle (Dugué et al., 2011; Haegens et al., 2011; Spaak et al., 2012).

As recent evidence contradicted direct connection between alpha activity and sensory information processing in early visual cortex (Zhigalov & Jensen, 2020), the objective of the current study is to investigate whether alpha modulation is connected to ‘gain control’ in early sensory areas through modulation of excitability (Foxe & Snyder, 2011; Jensen & Mazaheri, 2010; Van Diepen et al., 2019) or whether inhibitory effects are exhibited at later stages of the processing stream (Yang et al., 2023; Zhigalov & Jensen, 2020; Zumer et al., 2014), gating feedforward or feedback communication between sensory areas (Bauer et al., 2020; Haegens et al., 2015; Uemura et al., 2021).

To this end, we applied frequency-tagging, the rhythmic presentation of sensory stimuli, which elicits steady-state sensory evoked potentials or fields (SSEP/SSEF), consisting of rhythmic neuronal activity in the frequency of stimulation (Brickwedde et al., 2020; Colon et al., 2012; Dinse et al., 1998; Marzoll et al., 2018; Regan, 1982; Snyder, 1992; Stapells et al., 1984; Tobimatsu et al., 1999). The magnitude of SSEPs is attention-dependent (de Jong et al., 2010; Müller et al., 1998; Müller & Hillyard, 2000; Porcu et al., 2013; Saupe et al., 2009; Toffanin et al., 2009) even for frequencies too fast to perceive consciously (Brickwedde et al., 2022; Zhigalov et al., 2019). Their scalp topography reveals that SSEPs are most strongly observable over occipital areas for visual stimuli generated in the visual cortex, and over temporal (MEG) or fronto-to-central (EEG) areas for auditory stimuli generated in the auditory cortex (de Jong et al., 2010; Hari et al., 1989; Pantev et al., 1996; Regan, 1982). Their distinct response, localisation and attention-dependence provide an optimal tool to study sensory signal processing over time.

The aim of our initial EEG study was to directly investigate if the cue-induced modulation of alpha activity coincides with the suppression of frequency-tagging responses in task-irrelevant modalities. Based on previous studies, we utilized a cross-modal attention paradigm, in which symbolic visual cues signalled the target modality (visual or auditory) of an upcoming

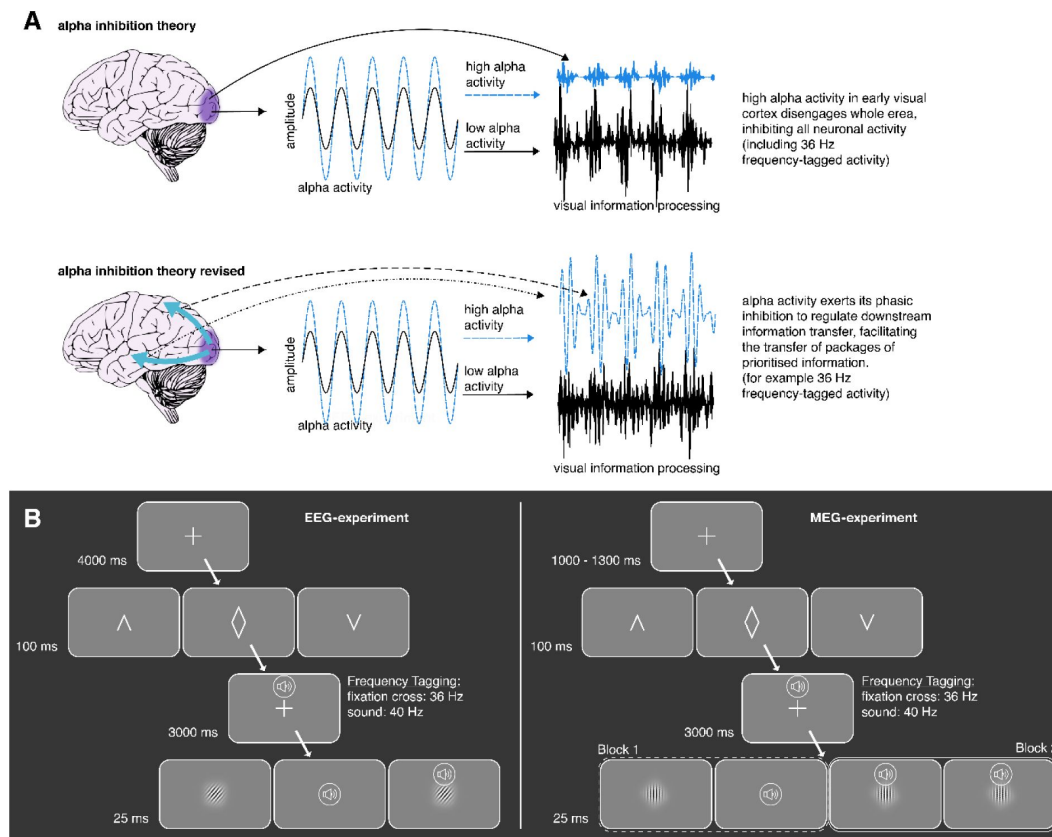


Figure 1.

Illustration the alpha inhibition theory.

A, the alpha inhibition theory suggests that alpha inhibits sensory information processing in a phasic manner. If alpha activity is high, it suggests that the whole area is inhibited and thereby disengaged (Foxy et al., 1998; Jensen & Mazaheri, 2010; Klimesch et al., 2007). We propose a revision of this theory, whereby alpha activity exerts its phasic inhibition to regulate downstream information transfer, creating enhanced signal packages of prioritised information (see also Yang et al., 2023; Zhigalov & Jensen, 2020; Zumer et al., 2014). **B, Illustration of the cross-model discrimination task.** LEFT: in the EEG-experiment, trials were separated by a 4 s interval, in which a fixation cross was displayed. A brief central presentation of the cue (100 ms) initiated the trial, signaling the target modality (see figure above from left to right: auditory, unspecified, visual). In the cue-to-target interval, the fixation cross was frequency-tagged at 36 Hz. At the same time, a sound was displayed over headphones, which was frequency-tagged at 40 Hz. Both tones and fixation cross contained no task-relevant information. The target consisting either of a Gabor patch or a tone was presented for 25 ms. Participants had to differentiate between 3 different Gabor rotations or tone pitches, respectively. In 50% of auditory and visually cued trials, a distractor in form of a random pitch or rotation of the un-cued modality was presented alongside the target. RIGHT: The MEG-experiment followed an almost identical setup. This time, trials were separated by a 1 s interval followed by a random jitter of 0 to 300 ms. The visual task was adjusted to now require discrimination between 3 different Gabor patch frequencies for the visual task. Lastly, a blocked design was incorporated where in block 1, no distractors were presented while in block 2, a random distractor from the stimulus pool of the non-target modality was always presented.

discrimination task (e.g., van Diepen et al., 2015 [↗](#)). Here, we included an additional experimental manipulation in the form of frequency-tagging to assess the involvement of the auditory and visual systems in the *cue-to-target interval*. We were also interested in the relationship between alpha modulation, SSEPs, and attentional performance on a trial-by-trial basis.

In line with previous results, we hypothesized that signalling an upcoming auditory target would lead to increased alpha activity over visual occipital regions as well as increased SSEP responses to auditory and decreased SSEP to visual stimuli as indexed by frequency-tagging. Furthermore, we aimed to explore whether there is a direct connection between alpha activity and frequency-tagging responses in other areas than primary visual cortex. In brief, while we observed the expected cue-induced alpha modulation in our initial EEG-study, the amplitude of auditory as well as *visual* SSEPs increased just prior to the onset of the auditory target.

To confirm our novel findings as well as analyse the cortical origin of the observed SSEP-changes, we replicated our study using MEG. Here we made slight adjustments to the experimental design by implementing a blocked design. The initial block involved no distractors, while the subsequent block consistently included a distractor. Furthermore, we modified the visual orientation task to align more closely with the auditory task.

The transition to a blocked design allows for a more controlled comparison between conditions with and without distractors, providing insights into how the presence of distractors influences the observed effects. Additionally, modifying the visual orientation task to align with the auditory task may enhance the comparability of the two modalities and potentially reveal more nuanced interactions between attentional processes and sensory processing across modalities. Here we not only replicated the observation that the amplitude of both *visual and auditory* SSEPs increased just prior to the onset of the auditory target, but we are also able to localise the sources of these activities.

Results

To assess audio-visual excitability in anticipation of either visual or auditory targets, we applied visual cues to signal the modality of the upcoming target (auditory, visual, or unspecified). In a three-second cue-to-target interval, we frequency-tagged the fixation cross at 36 Hz and played 40 Hz amplitude modulating white noise. Participants had to either discriminate between three different pitch sounds (auditory target) or three different Gabor patch orientations (visual target). If the target modality was cued, 50% of the trials were accompanied by a random distractor from the target pool of the opposing sensory modality (see **Fig. 1B** [↗](#)). In our subsequent MEG-study, we slightly adjusted the experiment into a blocked design. Specifically, the first block contained no distractors while in the second block, a distractor was always present. Additionally, we exchanged the visual orientation task to be more in line with the auditory task. Participants now had to distinguish between three different frequencies of the Gabor patch stripes. Prior to the experiment, difficulty of pitch sounds, and Gabor patch frequency were calibrated for each individual, ascertaining a success rate between 55% to 75%.

Behavioural performance

In the EEG-study, we found that accuracy differed significantly between conditions ($F_{(5,105)} = 44.16$; $p < .001$). Overall, participants were significantly less accurate in the auditory discrimination task ('overall auditory', $M = 79\%$ correct, $SD = 11.7$) than in the visual discrimination task ('overall visual', $M = 97\%$, $SD = 2.3$; see **Fig. 2A** [↗](#)), with the worst performance occurring when auditory targets were paired with visual distractors (auditory +: $M = 74\%$ correct, $SD = 13$). Our adjustments in the MEG-study streamlined performances to be more in line between auditory and visual conditions, especially in the second block ($F_{(3,75)} = 10.26$; $p < .01$). In block 1, participants were

significantly more accurate in the auditory ('block 1 auditory', $M = 84\%$ correct, $SD = 10.03$) compared to the visual task ('block 1 visual', $M = 70\%$, $SD = 10.61$; see [Fig. 2C](#)). However, in block 2, there was no observable difference between visual and auditory task accuracy ('block 2 auditory', $M = 77\%$ correct, $SD = 10.03$; ('block 2 visual', $M = 78\%$, $SD = 13.15$).

Reaction times yielded a similar pattern, with auditory reactions (overall auditory: $M = 662$ ms, $SD = 136$) being slower than for the visual task (overall visual: $M = 597$ ms, $SD = 130$; main effect over all conditions: $F_{(5,105)} = 27.47$; $p < .001$; see [Fig. 2B](#)). This was mostly driven by slow responses to auditory targets paired with distractors (auditory +: $M = 723$ ms). In the MEG-study, reaction times were comparable between conditions (overall: $M = 661$ ms, $SD = 76$), with the exception of responses to visual targets in block 1 (block 1 visual: $M = 726$ ms, $SD = 85$), which were significantly slower than all other reaction times (main effect condition: $F_{(3,75)} = 18.13$; $p < .001$; see [Fig. 2D](#)).

Cues and distractors were behaviourally relevant, as both attentional benefit and distractor cost were observable in our data (see [suppl Fig. 1](#)). Interestingly, auditory distractors reduced reaction times to visual targets, which could be explained by a generally faster processing of auditory targets (Jain et al., 2015), possibly probing faster responses in visual tasks (Naue et al., 2011). In the MEG-study, the distractor cost could not be differentiated from learning effects, due to the blocked design. Nonetheless, the previously observed pattern of the EEG study was replicated.

Cross-modal cues differentially modulated pre-target alpha activity

We conducted a time-frequency analysis of power in the cue-to-target interval and found a stronger amplitude increase from baseline for auditory compared to visual target conditions starting around 2 s before target onset ([Fig. 3A-B](#)). We also calculated the time course of alpha power changes using the Hilbert-transformation (8 – 12 Hz, [Fig. 3C-D](#)). Consistent with previous work (Mazaheri et al., 2014; van Diepen & Mazaheri, 2017), cluster permutation analysis conducted using the last two seconds before target onset revealed two clusters of difference in alpha power when expecting an auditory compared to a visual target. These effects corresponded to clusters extending from -1.84 to -0.64 s ($p = .004$) and -0.62 to 0 s ($p = .005$). We did not find significant effects for the ambivalent condition, which we then excluded from further analyses (see [suppl Fig. 2](#) for the data in this condition). In the MEG-experiment, in line with previous studies (van Diepen & Mazaheri, 2017), condition differences in alpha activity were only significant in block 2, where distractors in a different modality were presented. Therefore, we performed the same analyses as previously described for the EEG-study only for the second half of the experiment (see [Fig. 3E-G](#)), revealing a significant cluster from -1.47 to -1.18 s ($p = .034$). Applying roughly the same time-window (-1.5 and -2 s), we conducted a source localization, contrasting the two conditions with cluster permutation analysis. We found that condition differences were located in early visual areas with a stronger effect on the right compared to the left hemisphere ($p < .01$; peak spm coordinates: 41 -82 -19 mm, in the right lingual gyrus see [Fig. 3H](#)).

Cross-modal cues increased the amplitude of the frequency-tagged responses across both modalities

To assess the temporal development of frequency-tagging responses, steady-state potentials were calculated using data band-pass filtered around the tagging frequency. Neuronal responses to the 40 Hz auditory tagging were strongest over central areas ([Fig. 4A](#)) and 36 Hz responses were strongest over occipital areas ([Fig. 4B](#)). In accordance with the EEG-data, neuronal responses to 40 Hz auditory tagging measured with MEG were strongest over temporal areas ([Fig. 4C](#)) and 36 Hz responses were strongest over occipital areas ([Fig. 4D](#)). As expected, the auditory tagging

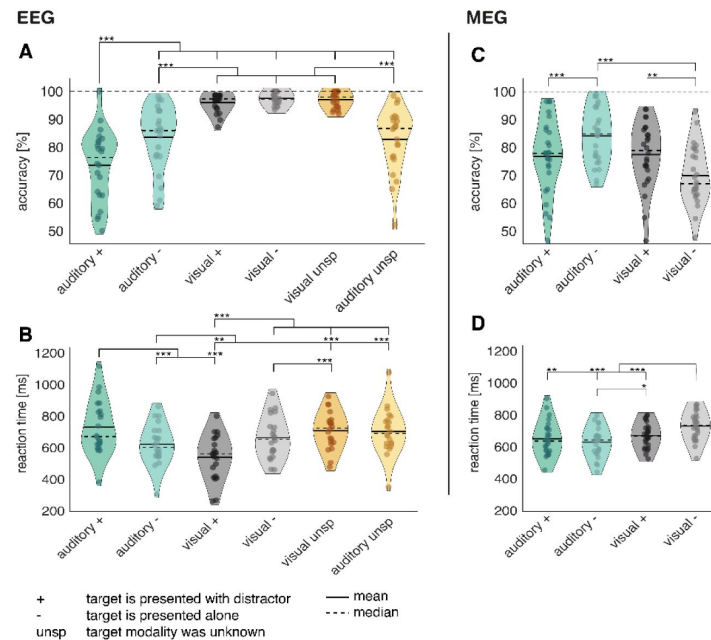


Figure 2.

Analysis of task accuracy and reaction time indicates increased difficulty of auditory targets in the EEG study, and comparable difficulties in the MEG study.

A, Task accuracy compared between all 6 experimental conditions reveals a drop in accuracy for responses to auditory targets. **B**, reaction times of correct trials compared between all 6 experimental conditions. The slowest reaction times are observable following auditory targets alongside visual distractors. **C**, Task accuracy differences were only observable in the first block without distractors. **D**, reaction times to visual targets in the first block were strongly decreased compared to all other conditions. In the second block, no significant difference in reaction times was observable. EEG Study: N = 22; MEG-Study: N = 27; *** sig. < .001; ** sig. < .01; * sig. < .05;

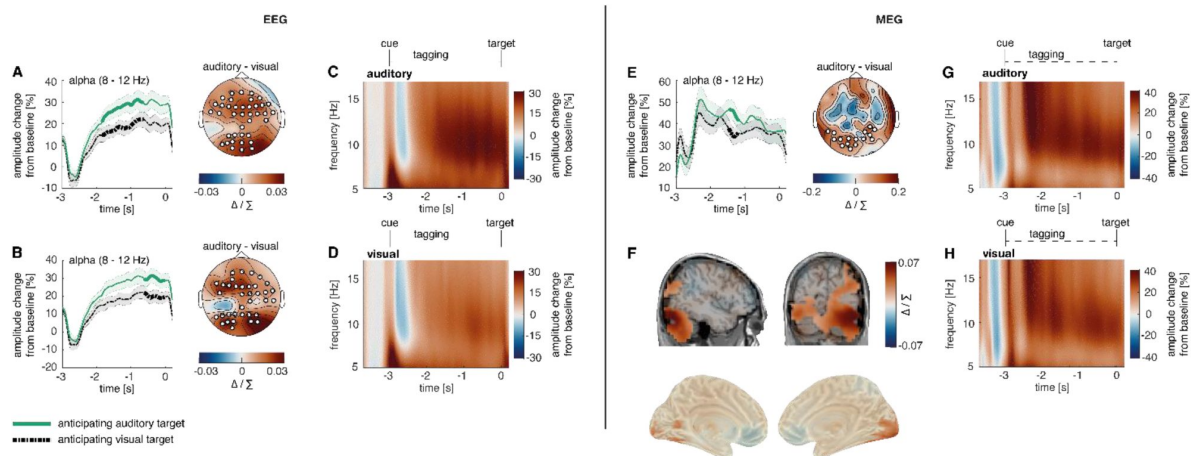


Figure 3.

Post-cue modality specific early visual modulation of alpha power in anticipation of an auditory versus a visual target

A-B, The time course of post-cue alpha power. Cluster permutation analysis resulted in two condition effects, both indicating heightened alpha activity when expecting an auditory compared to a visual target (C: $p < .01$; D: $p < .01$). **C-D,** Time-frequency representation of power in the cue-to-target interval. A greater increase in alpha power was observed when expecting an auditory target (average over significant electrodes for the condition difference in A). **E,** The time course of post-cue alpha power. Cluster permutation analysis resulted in a condition effects, indicating heightened alpha activity when expecting an auditory compared to a visual target ($p = .034$). **F,** source localization of the condition difference between expecting an auditory versus a visual target, revealing a significant cluster in early visual areas with stronger effects on the right hemisphere ($p < .01$). **G-H,** Time-frequency representation of power in the cue-to-target interval. A greater increase in alpha power was observed when expecting an auditory target (average over electrodes that showed maximal condition difference in E). **A, B, E,** Cluster electrodes are marked in white. Shading represents standard error from the mean; Δ / Σ represents (a-b)/(a+b) normalization.

response originated from the right-hemispheric early auditory cortex (cluster significance: $p < .01$; peak spm coordinates: 69 -22 5 mm, in the right superior temporal gyrus; see **Fig. 4E**) and the visual tagging response originated from the early visual cortex (cluster significance: $p < .001$; peak spm coordinates: 19 -105- 11 mm, in the right lingual gyrus; see **Fig. 4F**). Additionally, there was a significant reduced 40 Hz activity in the left-hemispheric visual-to-central cortex (cluster significance: $p < .01$).

To assess the differences between conditions, the Hilbert envelope of the steady-state potentials was analysed using cluster permutation analyses. When expecting an auditory target, the auditory 40 Hz frequency-tagging response was larger shortly before target onset (see **Fig. 4G**; -0.15 to -0.08 s; $p = .041$).

Surprisingly, the visual 36 Hz frequency-tagging response was likewise increased shortly before expecting an auditory compared to a visual target (see **Fig. 4H**; -0.16 to -0.06 s; $p = .014$). For both visual 36 Hz and auditory 40 Hz frequency-tagging responses, condition differences appeared strongest over mid-parietal regions. Applying the same analysis to MEG data over the last 500 ms before target onset replicated the EEG-results (see **Fig. 4I-J**; auditory target: $p = .043$; visual target: $p = .019$). As such, frequency tagging responses might reflect effort affecting the vigilance of the sensory system rather than the sensory-specific allocation of attention.

Source localization confirmed the condition difference in 36 Hz activity to originate from later stages of the processing stream, encompassing a wide range of areas, most strongly the medial occipital cortex, the calcarine fissure and the precuneus (cluster significance: $p = .047$; peak spm coordinates: 3 -66 44 mm, in the left and right precuneus; see **Fig. 4L**). In source space, the effect was not significant for 40 Hz activity ($p = .11$; peak spm coordinates: -9 -39 0 mm, in the left precuneus; see **Fig. 4K**).

Alpha power was positively correlated with amplitude of frequency tagged responses

Following the observation of condition differences in alpha activity and frequency-tagging responses, we were further interested in exploring whether these responses were connected. Accordingly, we conducted trial-by-trial correlations using alpha condition differences and their electrode positions as seed, which was correlated with frequency-tagging signals over all electrodes. Multiple comparison correction was applied by testing the correlation matrix against a zero-correlation matrix with a cluster permutation approach. A positive correlation was observed over right parietal-to-occipital areas between the late alpha cluster activity and both 40 Hz ($p = .009$) and 36 Hz ($p = .004$) frequency-tagging responses when expecting a visual target (see **Fig. 5A-B**). This result is further illustrated by a median split analysis between trials with high and low alpha power for each participant. It was highly significant for the visual 36 Hz response (**Fig. 5A**, middle columns, $p = .033$; $t_{(19)} = 2.29$; $BF_{(10)} = 1.91$) but did not reach significance for the visual 40 Hz response (**Fig. 5B**, middle column; $p = 0.20$; $t_{(19)} = 1.32$; $BF_{(10)} = 0.49$). Additionally, we averaged the correlation coefficient of each participant and calculated a one-sample t-test against 0. Both tests indicate strong correlations with alpha activity for both 40 Hz (**Fig. 5B**, right column; $p < .001$; $t_{(19)} = 4.95$; $BF_{(10)} = 306.93$) and 36 Hz activity (**Figure 6A**, right column; $p < .01$; $t_{(19)} = 3.66$; $BF_{(10)} = 23.57$). Applying the same analysis to the last 500 ms before target onset, we could replicate these results in our MEG data (see **Fig. 6F-G**; expecting a visual target, 36 Hz response: cluster significance: $p < .01$; median split: $p < .001$; $t_{(24)} = 4.33$; $BF_{(10)} = 127$; t-test: $p < .001$; $t_{(24)} = 5.33$; $BF_{(10)} = 1272$; expecting a visual target, 40 Hz response: cluster significance: $p < .001$; median split: $p < .001$; $t_{(24)} = 7.05$; $BF_{(10)} = 59443$; t-test: $p < .001$; $t_{(24)} = 6.75$; $BF_{(10)} = 30515$).

The same positive correlation with alpha activity was found when expecting an auditory target only for 36 Hz activity ($p = .031$), but not for 40 Hz activity (see **Fig. 6D**). A median split between high and low alpha activity ($p = .005$; $t_{(19)} = 3.14$; $BF_{(10)} = 8.53$) and correlation coefficients ($p =$

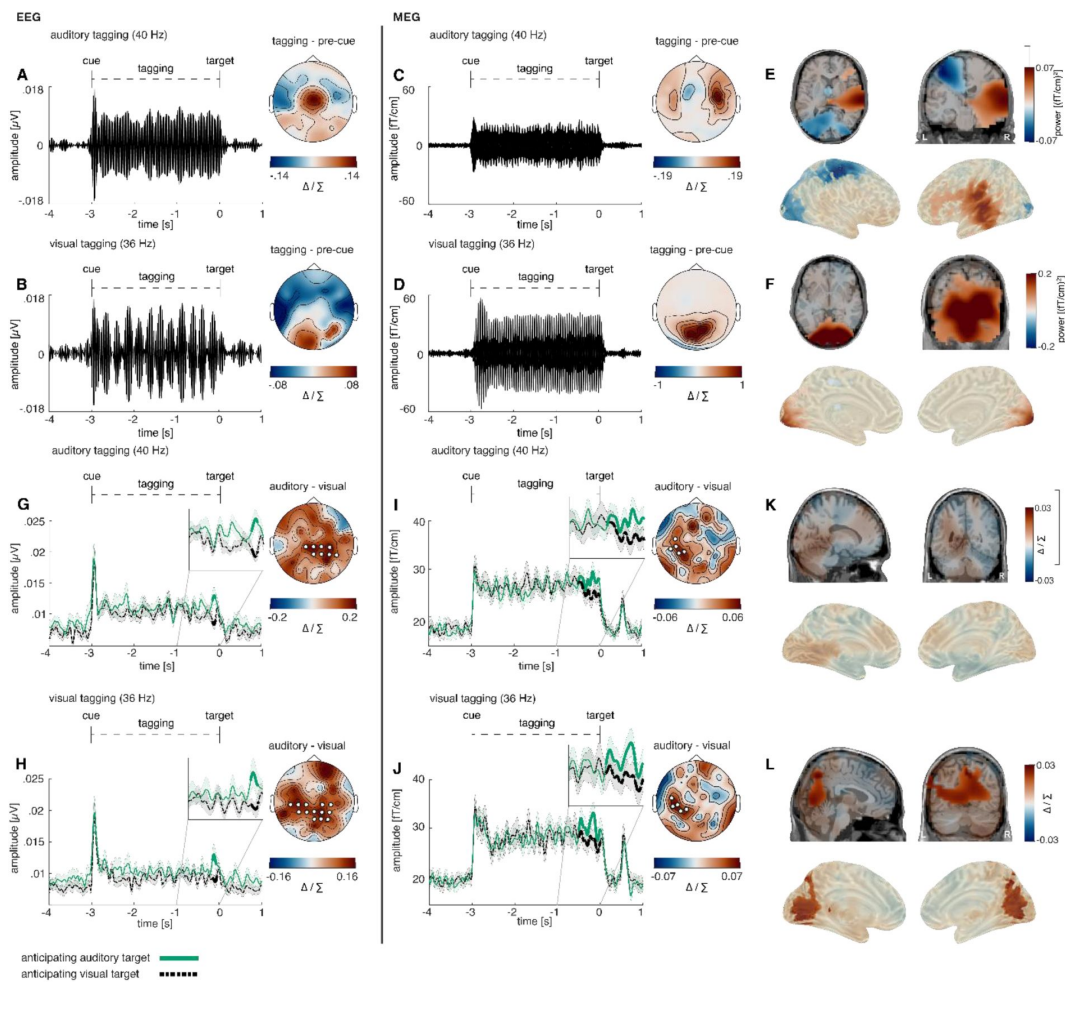


Figure 4.

Increase in amplitude of both visual and auditory frequency tagged responses when anticipating visual or auditory targets

Event-related potentials and scalp topographies reveal distinct modality specific responses at the tagged frequencies. **A)** auditory steady-state evoked potential (ASSEP) averaged over 6 central electrodes displaying the highest 40 Hz power (Fz, FC1, FC2, F11, F2, FCz). **B,** visual steady-state evoked potential (VSSEP) averaged over 4 occipital electrodes displaying the highest 36 Hz power (POz, O1, O2, Oz). **C,** auditory steady-state evoked fields (ASSEP) averaged over 20 temporal sensors displaying the highest 40 Hz power (10 right, 10 left). **D,** visual steady-state evoked fields (VSSEP) averaged over 10 occipital sensors, displaying the highest 36 Hz power. **E,** ASSEP source localization revealed a significant positive cluster in the right-hemispheric early auditory cortex ($p < .001$). **F,** VSSEP source localization revealed a significant positive cluster in the early visual cortex ($p < .001$). **G,I,** In both the EEG and the MEG study, the Hilbert-envelope of the 40 Hz ASSEP/ASSEP reveals an increase shortly before target onset when anticipating an auditory compared to a visual target (EEG: $p = .041$; MEG: $p = .043$); **H,J,** The Hilbert-envelope of the 36 Hz VSSEP/VSEP likewise reveals an increase shortly before target onset when anticipating an auditory compared to a visual target, both in the EEG as well as the MEG study (EEG: $p = .014$; MEG: $p = .019$). **K,** Condition differences in the 40 Hz ASSEP response did not reach significance in sensor space. **L,** Condition differences in the 36 Hz VSSEP response were significant over several areas of the visual stream, including most strongly the medial occipital cortex, the calcarine fissure, and the precuneus ($p = .047$); note: Cluster electrodes are marked in white. Shading represents standard error from the mean. Δ/Σ represents (a-b)/(a+b) normalization.

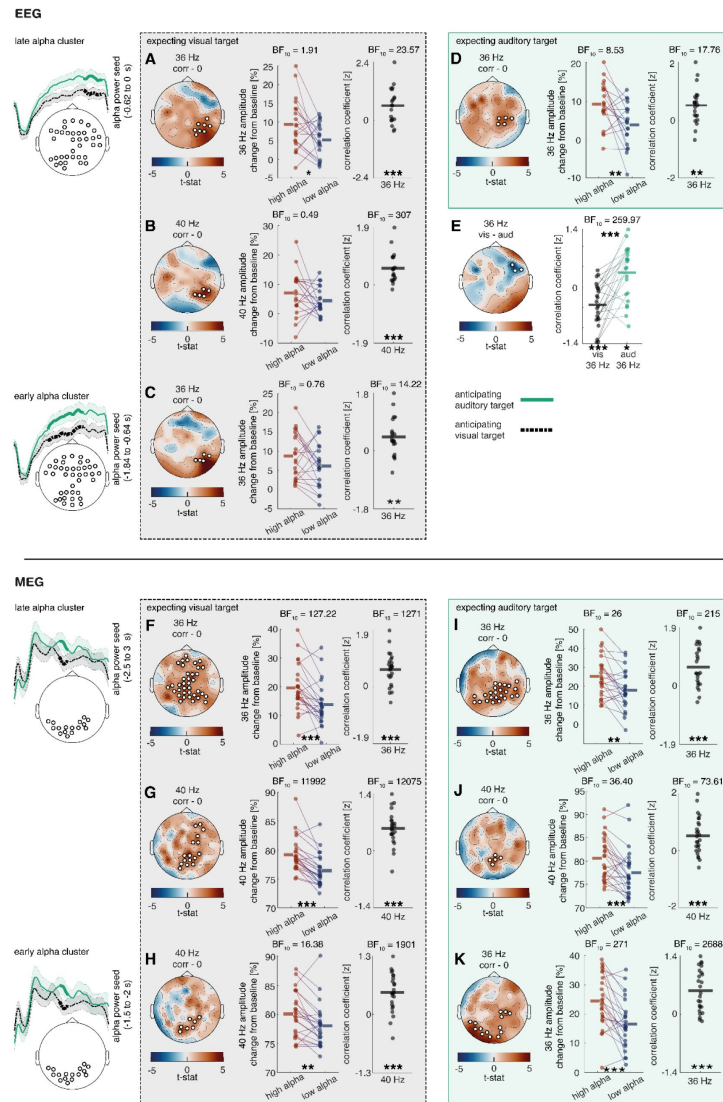


Figure 5.

Relationship between cue induced alpha modulation and amplitude of frequency tagged responses.

Previously obtained alpha clusters (see Fig. 3) were correlated over trials with 40 Hz and 36 Hz clusters (see Fig. 4), where alpha electrodes/sensors were applied as seeds. The analysis was performed using a cluster-permutation approach, testing a correlation model against a 0-correlation model. Clusters significantly diverging from the 0-correlation model are presented topographically. Additionally, median splits between high and low alpha trials as well as correlation coefficients of these clusters are displayed for all participants **A-B**, a positive correlation is visible between alpha activity in the last 400 ms and steady state potentials shortly before target onset when expecting a visual target (36 Hz: $p = .013$; 40 Hz: $p = .009$). **D**, when expecting an auditory target, there is a positive correlation visible between alpha activity in the last 400 ms and 36 Hz activity shortly before target onset ($p = .010$). **E**, the correlation between alpha activity 400 ms and 36 Hz activity shortly before target onset changes its direction depending on whether an auditory or a visual target is expected ($p = .037$). **C**, a positive correlation is also visible between alpha activity as early as ~1200 ms to 400 ms and 36 Hz activity shortly before target onset when expecting a visual target ($p = .016$). **F-H**, a positive correlation is visible between alpha activity in the last 500 ms as well as alpha activity in the last 1500ms-1000 ms and steady state potentials shortly before target onset when expecting a visual target (36 Hz late: $p = .013$; 40 Hz late: $p = .009$; 40 Hz early: $p = .002$). **I-K**, when expecting an auditory target, there is a positive correlation between alpha activity in the last 500 ms as well as alpha activity in the last 1500ms-1000 ms and steady state potentials shortly before target onset (36 Hz late: $p < .001$; 40 Hz late: $p = .005$; 36Hz early: $p = .011$). **A-K**, EEG: $N = 22$; MEG: $N = 27$; *** sig. $< .001$; ** sig. $< .01$; * sig. $< .05$. + sig. $< .1$

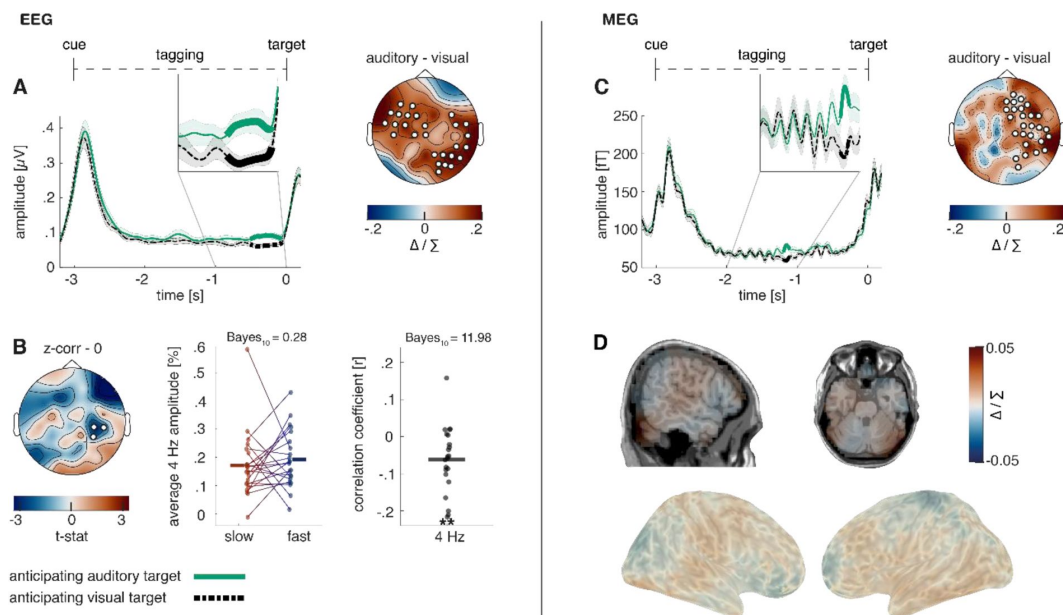


Figure 6.

Steady-state response in the intermodulation frequency and its behavioural relevance.

A, the Hilbert-envelope of the 4 Hz steady-state response reveals an increase shortly before target onset when anticipating an auditory compared to a visual target ($p < .01$). **B**, there is a trial-by-trial correlation between 4 Hz activity and reaction time when a visual target without distractor was presented. The correlation is further illustrated by a median split between fast and slow reaction time trials as well as by correlation coefficients for each participant. **C**, replication of the results presented in (A) in our MEG-study ($p = .006$). **D**, source localization showed activity over auditory sensory areas, but did not reach significance. EEG: $N = 22$; MEG: $N = 27$; ** sig. $< .01$;

.002; $t_{(19)} = 3.52$; $BF_{(10)} = 17.76$) provided moderate to strong evidence for this effect. In our MEG dataset, alpha activity 500 ms before auditory target onset correlated with both 36 Hz activity (see **Fig6. I-J**; cluster significance: $p < .001$; median split: $p = .001$; $t_{(23)} = 3.62$; $BF_{(10)} = 25.51$; t-test: $p < .001$; $t_{(23)} = 4.60$; $BF_{(10)} = 216$) and 40 Hz activity (cluster significance: $p = .005$; median split: $p < .001$; $t_{(25)} = 3.75$; $BF_{(10)} = 36.40$; t-test: $p < .001$; $t_{(25)} = 4.06$; $BF_{(10)} = 73.61$).

Additionally, we compared how correlation coefficients between alpha activity and frequency-tagging differed when anticipating an auditory versus a visual target. Multiple comparison correction was applied with cluster permutation analysis. Interestingly, an interaction between the strength of the correlation associating alpha and 36 Hz activity and condition became apparent ($p = .044$; see **Fig. 6E**) and was observed most strongly over right-central electrodes. Comparing the correlation coefficients of participants over this cluster revealed a strong effect and even a change of direction in the correlation ($p < .001$; $t_{(21)} = -4.76$; $BF_{(10)} = 259.97$). Particularly, when expecting a visual target, there was a negative correlation between 36 Hz and alpha activity, which turned positive when expecting an auditory target. Both correlations also differed significantly from 0 (expecting a visual target: $p < .001$; $t_{(21)} = -3.87$; $BF_{(10)} = 39.62$; expecting an auditory target: $p = .020$; $t_{(21)} = 2.53$; $BF_{(10)} = 2.83$). In contrast to the previously observed positive correlation between alpha activity and 36 Hz activity, the significant electrode cluster was located more ventrally. This effect could possibly hint at dynamic adaptability of oscillatory alpha effects on later processing stages. However, this could not be replicated in our MEG dataset.

It is further noteworthy that the correlation between alpha activity and 36 Hz frequency-tagging response when expecting a visual target, was also present when using the early alpha cluster as seed (~ 1200 to 400 ms before target onset, see **Fig. 6C**), in which case alpha activity preceded the 36 Hz activity ($p = .016$). For this correlation, the median split between high and low alpha trials did not reach significance ($p = .11$; $t_{(20)} = 1.69$; $BF_{(10)} = 0.76$). Testing correlation coefficients against 0 again revealed a significant effect ($p = .003$; $t_{(20)} = 3.39$; $BF_{(10)} = 14.22$). We could confirm these findings in our MEG-dataset, revealing a significant correlation between alpha activity during last 1 to 1.5 s before target onset and 36 Hz frequency-tagging response during the last 500 ms prior to an auditory target (see **Fig. 6K**; cluster significance: $p = .01$; median split: $p < .001$; $t_{(24)} = 4.66$; $BF_{(10)} = 271$; t-test: $p < .001$; $t_{(24)} = 5.66$; $BF_{(10)} = 2688$). The same alpha activity correlated with 40 Hz activity during the last 500 ms prior to a visual target (see **Fig. 6H**; cluster significance: $p = .002$; median split: $p = .002$; $t_{(23)} = 3.41$; $BF_{(10)} = 16.38$; t-test: $p < .001$; $t_{(23)} = 5.57$; $BF_{(10)} = 1901$).

Lastly, both alpha activity as well as 36 Hz frequency-tagging 500 ms before target onset activity correlated negatively with reaction time on a trial-by-trial basis, indicating faster reaction times in trials with higher pre-stimulus activity (alpha: $p = .037$; median split: $p = .013$; $t_{(25)} = -2.67$; $BF_{(10)} = 3.78$; t-test: $p < .01$; $t_{(25)} = -3.34$; $BF_{(10)} = 14.84$; 36 Hz: $p = .002$; median split: $p = .004$; $t_{(25)} = -3.20$; $BF_{(10)} = 8.98$; t-test: $p < .01$; $t_{(25)} = -3.46$; $BF_{(10)} = 19.12$. See **suppl Fig. 3-4**).

Intermodulation frequency

Lastly, we analysed the steady-state response of the intermodulation frequency at 4 Hz. Increased intermodulation shortly before target onset could be observed when expecting an auditory compared to a visual target (-0.51 to -0.0620 s; $p < .001$). This effect was strongest over left fronto-to-central electrodes and right central-to-occipital electrodes (see **Fig. 6A**). The same increase in the intermodulatory frequency could be observed in our second study, during the last 500 ms prior to target onset (see **Fig. 6C**; $p = .006$). In source space, a descriptive condition difference was visible in auditory sensory cortices, however this effect did not reach significance ($p = .49$; peak spm coordinates: 45 -83 -19 mm, in the right lingual gyrus; see **Fig. 6D**). With the goal to examine whether there are any behaviourally effects of the condition difference, trial by trial correlations with reaction times of each of the 2 auditory and visual conditions were performed. Only in the easiest condition, when expecting a visual target that was not accompanied by a distractor, a negative correlation with reaction time could be found, strongest over right central

electrodes (see **Fig. 6B**; $p = .046$). While a median split between slow and fast trials did not reach significance ($p = .50$; $t_{(21)} = -0.69$; $BF_{(10)} = .28$), testing the correlation coefficients against 0 revealed strong evidence for a correlation ($p = .004$; $t_{(21)} = -3.28$; $BF_{(10)} = 11.98$).

Discussion

The neuropsychological account of attention defines it as the selective facilitation (i.e., prioritization) of relevant sensory input and suppression of irrelevant sensory input. Oscillatory activity in the alpha range (~10 Hz) has been suggested to play a mechanistic role in attention through inhibition of irrelevant cortices, commonly referred to as the ‘alpha inhibition hypothesis’ (Foxe et al., 1998; Jensen & Mazaheri, 2010; Klimesch et al., 2007). In the current cross-modal attention study we directly tested this hypothesis by using frequency-tagging to specifically examine how cues signalling the modality of an upcoming target (either the auditory or visual modality) affected the responsiveness of the relevant and irrelevant sensory cortices prior to target onset. In-line with previous work, we observed a post-cue increase in posterior alpha power in anticipation of processing auditory targets. However, contrary to prevalent theories proposing visual gain suppression when focusing on the auditory modality, we observed that the amplitude of visual frequency-tagging responses increased just prior to the onset of the auditory target. This suggests that responsiveness of the visual stream was not inhibited when attention was directed to auditory processing and was not inhibited by occipital alpha activity. Our results reconcile previously paradoxical results on audio-visual attention and support the view that alpha activity gates downstream communication pathways.

Frequency-tagging

In the current experiment, we specifically chose to analyse the cue-to-target interval, where both visual and auditory SSEPs/SSEFs present preparatory states for the upcoming task, independent of task-related processing or performance. The magnitude of auditory SSEPs/SSVEFs was increased shortly before target onset when expecting a demanding auditory target compared to a visual target, very much in line with previous reports (e.g., Saupe et al., 2009). In contrast to the results reported in Saupe et al., (2009), where visual SSEPs decreased when attending the auditory modality, visual SSEPs/SSVEFs increased shortly before target onset when expecting an auditory target in our data. This is especially surprising as auditory targets were frequently or in case of our second study, always accompanied by visual distractors, rendering it optimal for task success to completely ignore any visual input. As auditory targets were significantly more difficult than visual targets in our first study and of comparable difficulty in our second study, these results strongly speak to a vigilance increase of sensory processing independent of modality and an inability to selectively disengage one sensory modality in anticipation of a demanding task. This view is consistent with previous work in which visual SSEPs elicited by irrelevant background stimulation increased with task load in an auditory discrimination task (Jacoby et al., 2012). Furthermore, our results indicate that task demand is a strong candidate to reconcile previously seemingly paradox results, as splitting attention between the auditory and visual system seemed possible in simpler tasks (Driver & Spence, 1998; Saupe et al., 2009) and impossible under high demand (de Jong et al., 2010; Driver, 1996; Driver & Spence, 1998; Spence & Driver, 1996). An alternative account for our findings stems from the evidence, that participants are more likely to only perceive and react to the visual modality, when confronted with audio-visual stimuli (Colavita, 1974; Spence, 2009). However, this effect was mostly limited to speeded modality discrimination/target detection tasks (Sinnett et al., 2008; Spence, 2009). Furthermore, the increased difficulty of the here-used auditory stimuli was confirmed in a previous block-design study (van Diepen & Mazaheri, 2017) and in our second study, performances over the visual and auditory tasks were comparable. Nevertheless, visual dominance could play a role for auditory target difficulty as well as predictions over the reciprocity of the audio-visual relationship.

A revision of the Alpha inhibition hypothesis

Top-down cued changes in alpha power have now been widely viewed to play a functional role in directing attention: the processing of irrelevant information is attenuated by increasing alpha power in cortices involved with processing this information (Foxe, Simpson, & Ahlfors, 1998; Hanslmayr et al., 2007; Jensen & Mazaheri, 2010). However, recent evidence suggests that alpha activity does not inhibit gain in early sensory processing stages (Antonov et al., 2020; Gundlach et al., 2020; Gutteling et al., 2022; Zhigalov & Jensen, 2020). To date there has been no direct investigation into the effect of alpha increases on later stages in the processing stream. In the current study, as expected, we observed a post-cue increase in occipital alpha activity in anticipation of an auditory target. However, we also observed an increase in the amplitude of visual SSEPs during the cue-target interval, which directly contradicts the widespread view of alpha activity exerting ‘gain control’ in early sensory areas by regulating excitability (Foxe & Snyder, 2011; Jensen & Mazaheri, 2010; Van Diepen et al., 2019). Here we propose that alpha activity, rather than modulating early primary sensory processing, exhibits its inhibitory effects at later stages of the processing stream (Antonov et al., 2020; Gundlach et al., 2020; Zhigalov & Jensen, 2020; Zumer et al., 2014), gating feedforward or feedback communication between sensory areas (Bauer et al., 2020; Haegens et al., 2015; Uemura et al., 2021). Our data provides evidence in favour of this view, as we can show that alpha activity covaries over trials with SSEP magnitude in adjacent areas. If alpha activity exerted gain control in early visual regions, increased alpha activity would have to lead to a decrease in SSEP responses. In contrast, we observe that increased alpha activity originating from early visual cortex is related to *enhanced* visual processing at later stages of the processing stream, which we could confirm using source analysis. It seems plausible to assume that inhibition of other task-irrelevant communication pathways leads to prioritised and thereby enhanced processing over relevant pathways. In line with previous literature, we therefore suggest that alpha activity limits task-irrelevant feedforward communication, thereby enhancing processing capabilities in relevant downstream areas (see Fig. 1A). Furthermore, we could show that the magnitude of the correlation between alpha power and visual information processing varied between conditions, suggesting a dynamic and adaptive network.

It is known that the localisation of alpha activity reflects the retinotopic organisation of visual spatial attention topographically over the parietooccipital cortex (Kelly et al., 2006; Popov et al., 2019). Notably, recent studies provided evidence, that the same organisation can be observed for auditory attention. Specifically, the localisation of visual alpha activity in the parietooccipital cortex reflects the spatial direction of auditory attention (Klatt et al., 2021; Popov et al., 2021). This observation can be explained through micro-saccades towards the spatial location of sounds, which are irrevocably connected to alpha oscillations (Popov et al., 2021). While we did not manipulate spatial attention, our results fit well to the notion of visual alpha activity serving as a sensory orientation system, relaying visual information to task-relevant downstream processing areas, and blocking communication to irrelevant pathways.

The intermodulation frequency

Previous research showed that simultaneous frequency-tagging in multiple frequencies evokes a response in the intermodulation frequency ($f_1 - f_2$). In multimodal settings, this frequency is thought to reflect cross-modal integration (Drijvers et al., 2021). This is very well in line with our findings, where increased vigilance of the sensory system arising from anticipation of a difficult auditory target resulted in an increase in the intermodulatory frequency. Furthermore, we could show that this frequency covaries over trials with reaction time in the easiest condition, where visual targets were presented without any distractors. A lack of this connection in other conditions might reflect increasing interferences from higher task difficulty, rather than a lack of the effect itself, but this remains to be tested. We cannot exclude an alternative explanation, as theta

oscillations are known to be involved in movement preparation, it is possible that phase-resets could lead to time-locked appearance of these oscillations (Lakatos et al., 2008 [↗](#); Tomassini et al., 2017 [↗](#)).

Conclusion

Our results taken together suggest that under high task difficulty, audio-visual excitability is enhanced, reflecting an increase in vigilance for the sensory system, even if this increases processing of distracting information. We showed that this vigilance shift, as reflected by SSEP/SSEF responses, is regulated by alpha activity, presumably through relaying of sensory information over communication pathways, thereby controlling the downstream flow of sensory information.

Materials and Methods

Participants EEG-Study

In total, 24 healthy volunteers participated in this study (mean age: 19.1 ± 1.8 SD; 17 women). Due to technical difficulties, one participant could not finish the experiment and one participant did not exceed chance level in the behavioural task ($\sim 33\%$). Both were therefore removed from any further analysis. All remaining participants reported normal or corrected-to-normal vision, no history of psychiatric or neurological illness and provided written informed consent. After completion of the experiment, participants received either monetary compensation or certification of their participation for their university course program. The study protocol was approved by the Ethics Committee of the School of Psychology at the University of Birmingham and is in accordance with the Declaration of Helsinki.

Participants MEG-Study

In total, 28 healthy volunteers participated in this study (mean age: 23.4 ± 3.6 SD; 20 women). One participant was removed from further analysis, as they only responded to $\sim 42\%$ of trials correctly in the second block, which related to 27/19 trials per condition respectively. All remaining participants reported normal or corrected-to-normal vision, no history of psychiatric or neurological illness and provided written informed consent. After completion of the experiment, participants received either monetary compensation or certification of their participation for their university course program. The study protocol was approved by the Ethics Committee of the School of Psychology at the University of Birmingham and is in accordance with the Declaration of Helsinki.

Cross-modal attention paradigm EEG-Study

Each trial was initiated by a brief presentation of a cue (100 ms) signalling the modality of the upcoming discrimination task (v-shape: visual modality; inversed v-shape: auditory modality; diamond-shape: unspecified). During the following cue-to-target interval, the fixation cross was frequency-tagged at 36 Hz. At the same time, a 40 Hz frequency-tagged sound (amplitude modulated white noise) was played over headphones (see [Fig. 1](#) [↗](#)). The volume of tones was initially adjusted to a level that was clearly perceivable but not uncomfortable and remained stable over participants. No task was connected to this interval and participants did not need to pay attention to either the sound or the fixation cross. After three seconds, the frequency-tagging stopped, and right after the cessation of stimuli, the target was presented for a very brief moment (25 ms). It consisted either of a Gabor patch (visual modality) or a sound (auditory modality). If the target modality was visual, participants had to use the three arrow buttons on the keyboard to indicate whether the Gabor patch was tilted to the left (-10° ; left arrow button), vertical (0° ; down arrow button), or tilted to the right (10° ; right arrow button). Additionally, in 50% of the trials, a

random distractor from the pool of auditory targets was presented simultaneously to the visual target over headphones. Similarly, if the target modality was auditory, participants used the same buttons to indicate whether the pitch of the tone was low (500 Hz), medium (1000 Hz) or high (2000 Hz). Again, in 50% of the trials, a random distractor from the pool of visual targets was also presented. If the cue was unspecified (diamond shape), either a visual (50% of unspecified trials) or an auditory target was presented, never accompanied by any distractors. Experimental trials were separated by an inter-trial interval of 4 seconds, to avoid carry-over effects from previous trials. The resulting 6 conditions were randomly ordered and balanced out over the experiment.

During the experiment, participants were instructed to keep their gaze locked to a fixation cross presented at the centre of the screen. Preceding data collection, participants performed 36 practice trials to get accustomed to the task and the target stimuli. The ensuing experiment was split into 26 trial sequences, separated by self-chosen breaks, which together resulted in 468 trials and lasted between 80 and 90 minutes. The discrimination task was programmed and presented with MATLAB® R2020b and Psychtoolbox-3 on an LCD-monitor featuring a 140 Hz refresh rate. The onset of the visual and auditory tagging frequencies (i.e. steady state stimuli) were tracked using the Cedrus Stimtracker (<https://cedrus.com/stimtracker/index.htm> )

Adjustments to the attention paradigm in the MEG-Study

In our second study, we removed all ambiguity concerning targets and distractors and therefore developed a blocked design, incorporating two blocks. The first block did not display distractors and only correctly predicting target cues were presented. Cues in the second block were likewise always correctly indicating the target modality, but this time, each target was accompanied by a random distractor from the non-target modality.

Furthermore, the visual task was adjusted to be more in line with the auditory task. As such, the Gabor patches now featured stripes in different frequencies (e.g. a low number of stripes, a medium number of stripes and a high number of stripes). The participant's task was to discriminate between these three Gabor patches. As auditory targets had been markedly more difficult in our first study, we now included a brief difficulty calibration prior to the experiment. First, we presented 21 Gabor patches with 3 different amounts of stripes following a standard difficulty. If participants could discriminate them correctly 55 – 75% of the time, this difficulty setting was chosen. Otherwise, depending on the performance, the stripe-frequency of the Gabor patches was adjusted. There were maximally 3 sessions of 21 Gabor patches, after which we had enough data to calibrate the individual difficulty setting.

The same procedure was then performed with the difficulty of the tones, calibrating the pitch frequency for each individual participant.

Lastly, visual frequency-tagging stimulation now followed a sinusoidal contrast-change rather than an on-off stimulation, which was possible due to a high-resolution projector featuring a refresh rate of 1440 Hz (PROPixx DLP LED projector ;VPixx Technologies Inc., Canada).

Eye-tracking

To make sure participants focused on the fixation cross during the cue-to-target interval, we incorporated eye-tracking into our MEG-experiment (EyeLink 1000 Plus). Correct trials of the second block were analysed for vertical and horizontal eye-movements. To remove blinks, trials with very large eye-movements (> 10 degrees of visual angle) were removed from the data (See [suppl Fig. 5](#) )

Behavioral analysis

We were interested in accuracy in the discrimination of visual targets and auditory targets, as well as reaction times. Furthermore, we examined the distraction cost of having a target presented with a distractor of a different modality as well as the reaction time to make the target discrimination. The distraction cost was calculated as the reaction time difference between cued targets with distractors (i.e. visual and auditory stimuli presented together) and cued targets without distractors (either a visual or auditory stimulus presented alone). All incorrect trials as well as trials with reaction times faster than 100 ms or exceeding 1500 ms were removed from analysis (0.5 % too fast, 9.2% too slow).

EEG data acquisition

All EEG recordings were conducted using a WaveGuard Cap (ANTneuro), featuring 64 Ag/AgCL electrodes (10-10 system; ground: Fz; reference: Cpz; EOG: left canthus). Electrodes positions were prepared with OneStep cleargel conductive paste and impedances were kept below 100 k Ω . The measured signal was transmitted using an ANTneuro EEGosports amplifier (low-pass filter: 150 Hz; high-pass filter: 0.5 Hz; sampling rate: 500 Hz).

MEG data acquisition

Prior to the experiment, fiducial positions and head-shape were recorded using a FASTRAK system (Polhemus, USA). The experiment took place in a dimly lit room, where participants were seated in a comfortable chair in the gantry of a 306-sensor TRIUX Elekta system with 204 orthogonal planar gradiometers and 102 magnetometers (Elekta, Finland). The 71*40 cm screen was positioned at ~1.40 m distance from the participant.

EEG Preprocessing

Offline analyses were performed in MATLAB ® R2020b. The data was pruned from artifacts by visual inspection using the EEGLAB toolbox (Delorme & Makeig, 2004). Additionally, blinks and ocular artefacts were removed from the data using independent component analysis (ICA). EEG channels were re-referenced to an average of all channels (excluding EOG).

MEG Preprocessing

Offline analyses were performed in MATLAB ® R2020b and Python. Spatiotemporal Signal-Source-Separation (SSS) was applied to the raw data via MNE's inbuilt maxfilter function with a duration window of 10 s and a correlation value of .9. The data was pruned from artifacts by visual inspection using the Fieldtrip toolbox (Delorme & Makeig, 2004). Additionally, blinks and ocular artefacts were removed from the data using independent component analysis (ICA). In sensor space, planar gradiometers were combined for further analyses. In source space, all individual planar gradiometers were analysed.

Amplitude of the evoked frequency-tagging response

To investigate the temporal dynamics of amplitude of the frequency tagged responses after the onset of the attentional cues (also the precise onset of the frequency tagged stimuli) the data was epoched into 6-second segments starting 1.5 seconds prior to cue onset. Next the data were narrow-band filtered around the 36 Hz activity to capture the visual frequency-tagging, 40 Hz activity in to capture the auditory frequency-tagging, and the intermodulation frequency at 4 Hz, which can be derived by subtracting both frequency-tagging responses ($f_i = f_{\text{auditory}} - f_{\text{visual}}$; see Drijvers, Spaak & Jensen, 2020). Here we used a Blackmann-windowed sinc filters adapted to a suitable ratio of temporal and frequency resolution for the specific frequency of each of the tagged signals: filter order 116 for 35.5 to 36.5 Hz and 39.5 to 40.5, filter order 344 for 3.5 to 4.5 Hz. The filtered data at each of the tagged frequencies as well the intermodulated frequency were baseline

corrected (interval between 700 and 200 ms preceding cue onset) before calculating the average over trials to obtain steady-state evoked potentials). The power envelope of the SSEPs of tagged frequencies was estimated using Hilbert transformation.

Temporal dynamics of the induced EEG changes

In addition to looking at the cue evoked changes in the amplitude of the frequency tagged signals, we investigated the induced changes in the EEG signal at the frequencies of the tagged auditory and visual stimuli, alpha activity (9-11 Hz, filter order 276 for 7.5 to 12.5 Hz), as well as the intermodulation frequency (4 Hz; filter order 344 for 3.5 to 4.5 Hz). Here rather than averaging the epoched data filtered at the specific frequency ranges, we performed the Hilbert transform, and averaged the power-envelope of the specific frequencies across trials. This approach is very much analogous to the standard time-frequency analysis using convolutions (van Diepen & Mazaheri, 2017 [↗](#); Zhigalov et al., 2019 [↗](#)), but affords more control concerning temporal versus frequency resolution to examine the temporal dynamics of the specific frequencies of interest. In our second study, the individual peak alpha frequency was used for bandpass-filtering in contrast to a standardised band applied in the first study.

Time-frequency representations of power

In addition to the estimating frequency power envelopes, Time-frequency representations (TFRs) of power of the EEG signal were estimated using the Fieldtrip toolbox (Oostenveld et al., 2011). The power or frequencies between 5 and 20 Hz were calculated for each trial, using a sliding time window (frequency steps: 0.5; time steps: 10 ms). The length of the window was adjusted to a length of 3 cycles per frequency and tapered with a Hanning window. For each trial, both datasets were normalised to display relative percent change from baseline using the following formula: $[(\text{activity} - \text{baseline}) / \text{baseline}]$, where baseline refers to the interval between 700 and 200 ms before cue onset. To estimate the topographical distribution of voltage differences between conditions, uncorrected power values were normalised applying the following formula: $[\Delta / \Sigma = (a - b) / (a + b)]$, where a and b reflect the different conditions.

Source localization

Source localization was performed with a beamformer approach using the Fieldtrip toolbox. Headmodels were created based on individual T1-scans fitted to fiducial points and head shapes. These data were fit to a 5mm 3d sourcemodel and warped into MNI-space. Two participants were missing individual T1-scans. In these cases, we applied a standardized T1-scan using the Colin 27 Average Brain Model (Holmes et al., 1998 [↗](#)). Frequency-domain data was localized using the Dynamic Imaging of Coherent Sources (DICS beamformer) method with a dpss taper of 2 Hz assuming fixed orientation. As condition differences between frequency-tagging responses were better estimated in the time-domain, we assessed them applying the Synthetic Aperture Magnetometry (SAM-beamformer) method with optimal fixed rotation (Sekihara et al., 2004 [↗](#)). Significant brain areas and peak coordinates were related to brain areas using the Anatomical Automatic Labeling (AAL) atlas for SPM8 (Tzourio-Mazoyer et al., 2002 [↗](#)). After statistical analysis, source localized data was interpolated onto the Colin 27 Average Brain Model MRI. Cerebellar and brainstem interpolations were excluded from the coordinate system.

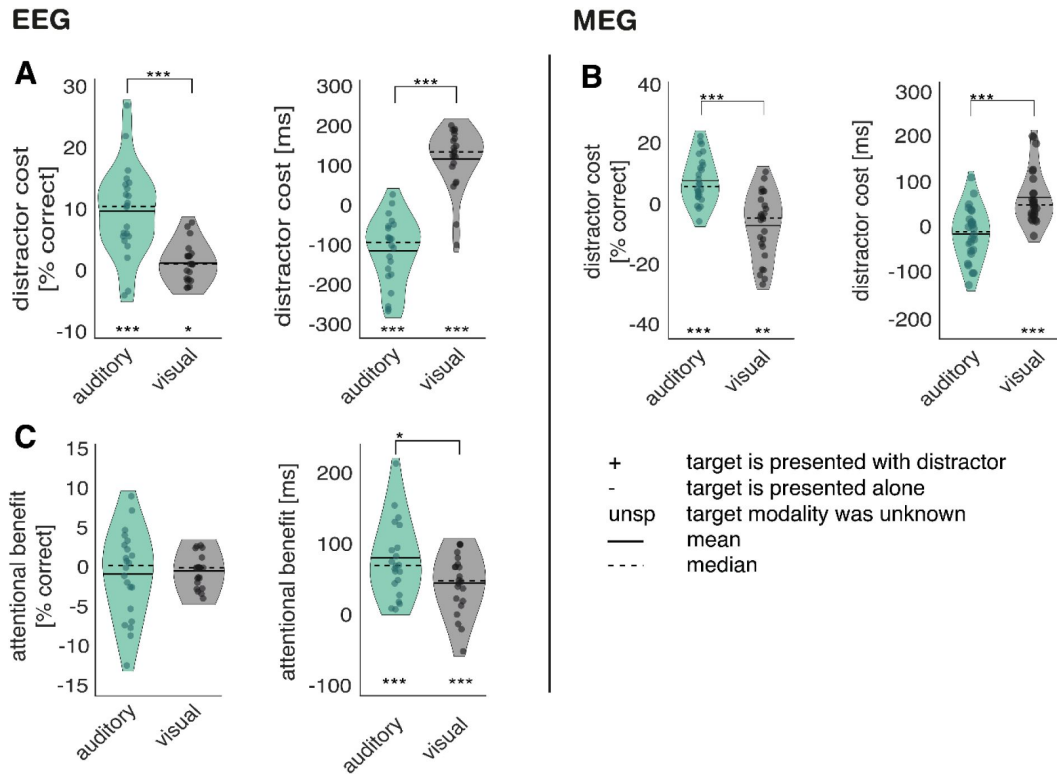
Statistical analysis

Condition differences in the behavioural task were estimated with paired t-tests and repeated measures ANOVAS, utilizing Tukey-Kramer Post-Hoc test. Power differences between conditions, as well as source-space contrasts were analysed using cluster permutation analysis (Maris & Oostenveld, 2007 [↗](#)). In this procedure, condition labels were randomly shuffled 1000 times, creating pairs of surrogate conditions. To test for significance, paired t-tests were conducted for each data point and each channel, resulting in one t-matrix for real conditions and 1000 t-matrices for surrogate conditions. Significant t-values ($p < .05$) were defined as clusters if there was at least

one significant data point present at the same time and frequency in at least two neighbouring channels. To correct for multiple comparisons, a condition difference was only assumed, if the maximum sum of t-values in a real cluster exceeded the same sum of 95% of the clusters found in the surrogate data. To replicate our results in the second study, we applied the same statistics and averaged the previously found time-intervals into windows of 500ms (e.g.: if we found an effect -0.51 to -0.0620 s prior to target onset we tested this effect for a time window of -0.5 to 0 s prior to target onset).

The relationship between induced changes in alpha activity and frequency tagged responses was assessed using trial by trial Spearman correlations. For each participant and each electrode, a correlation coefficient was calculated between the average activity in a previously identified cluster, which was used as seed (e.g., condition differences in alpha activity), and the average activity of the electrophysiological correlate of interest (e.g., 36 Hz activity over the previously identified time window). The correlation coefficients were z-transformed, and the resulting channel by participant matrix was tested against null-correlation model using the cluster permutation approach described above. Derived clusters were additionally tested and visualised by comparing median split trials of high vs low activity. For this analysis, outliers (values deviating more than 2 standard deviations from the mean) were excluded. Furthermore, the average correlation coefficient of the cluster was tested against a 0-correlation model for each participant using t-statistics. Lastly, an interaction between electrophysiological correlations and conditions was performed using correlation coefficients for each participant and electrode, testing them between conditions using the cluster permutation approach. Perceptually uniform and universally readable colormaps were applied to all visualisations (Cramer et al., 2020 [DOI](#)). All data are presented as mean $\pm$ standard error of the mean (SEM).

Supplementary Materials



SUPPL Fig. 1

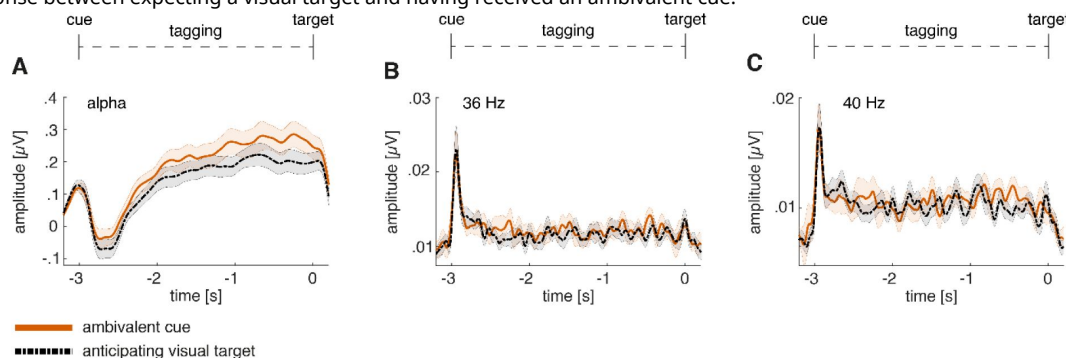
Distractor cost and attentional benefit.

A-B, Illustration of distractor cost: mean performance over trials with distractors was subtracted from mean performance over trials without distractors. Distractor effects were observable for accuracy as well as reaction time; **A**, accuracy: auditory- - auditory +: $M = 10.0\%$; $SD = 7.3$; $p < .001$; $t_{(21)} = 7.32$; visual- - visual+: $M = 1.5\%$; $SD = 3.06$; $p = .02$. The effect was stronger for auditory than for visual target trials ($p < .001$; $t_{(21)} = 7.67$). Reaction time: (auditory- - auditory+: $M = -108.1$ ms; $SD = 84.8$; $p < .001$; $t_{(21)} = -5.98$; visual- - visual+: $M = 123.6$ ms; $SD = 76.3$; $p < .001$; $t_{(21)} = 7.60$). auditory distractors decreased response time to visual targets ($p < .001$; $t_{(21)} = -11.99$). **B**, (accuracy: auditory- - auditory +: $M = 7.2\%$; $SD = 7.5$; $p = .001$; $t_{(25)} = 4.9$; visual- - visual+: $M = -7.6\%$; $SD = 10.80$; $p < .01$; $t_{(25)} = -3.59$; Reaction time: auditory- - auditory +: $M = -20.64$ ms; $SD = 57.6$; n.s.: $p = .08$; $t_{(25)} = -1.83$; visual- - visual+: $M = 60.1$ ms; $SD = 58.52$; $p < .001$; $t_{(25)} = 5.23$). **C**, Illustration of attentional benefit: mean performance over unspecified trials was subtracted from mean performance over modality-cued trials without distractor. attentional benefit auditory: unspecifically cued auditory targets - informatively cued auditory targets = $M = 81.2$ ms; $SD = 54.9$; $p < .001$; $t_{(21)} = 6.94$; attentional benefit visual: unspecifically cued visual targets - informatively cued visual targets -; $M = 54.4$ ms; $SD = 41.1$; $p < .001$; $t_{(21)} = 5.19$). The magnitude of the effect on reaction time also differed between conditions ($p = .043$; $t_{(21)} = 2.16$), with stronger attentional benefit for auditory target cues. Attentional cues did not affect response accuracy, neither in auditory nor visual target conditions (auditory: $p = 0.49$; visual: $p = 0.32$). EEG Study: $N = 22$; MEG-Study: $N = 27$; *** sig. $< .001$; ** sig. $< .01$; * sig. $< .05$;

SUPPL Fig. 2

Timecourse of alpha activity and frequency-tagging responses for the ambivalent compared to the visually-cued condition.

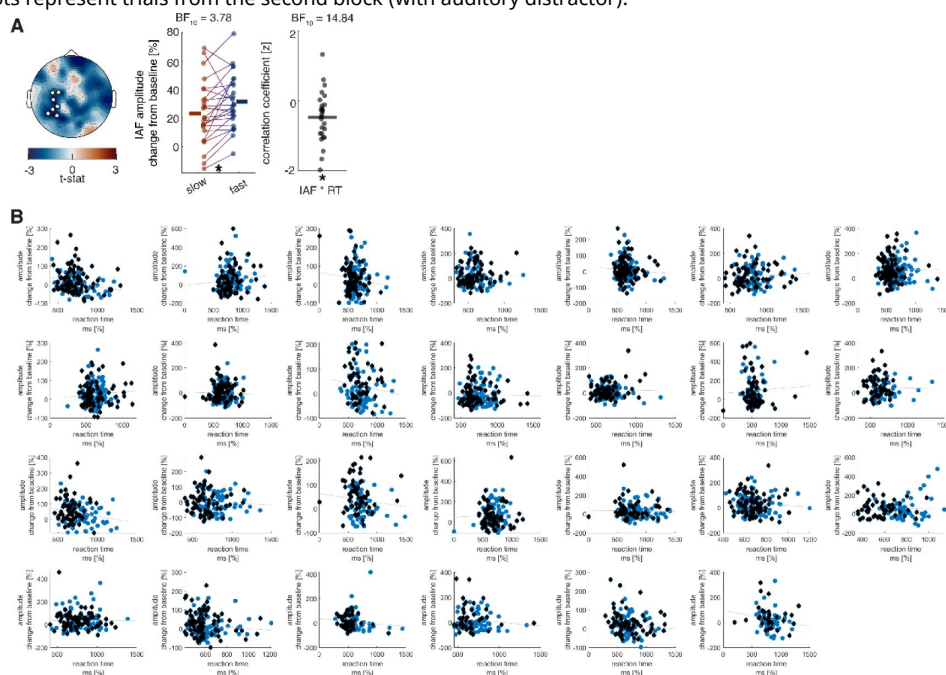
A, alpha activity compared between expecting a visual target and having received an ambivalent cue. **B**, 36 Hz frequency-tagging response between expecting a visual target and having received an ambivalent cue. **C**, 40 Hz frequency-tagging response between expecting a visual target and having received an ambivalent cue.

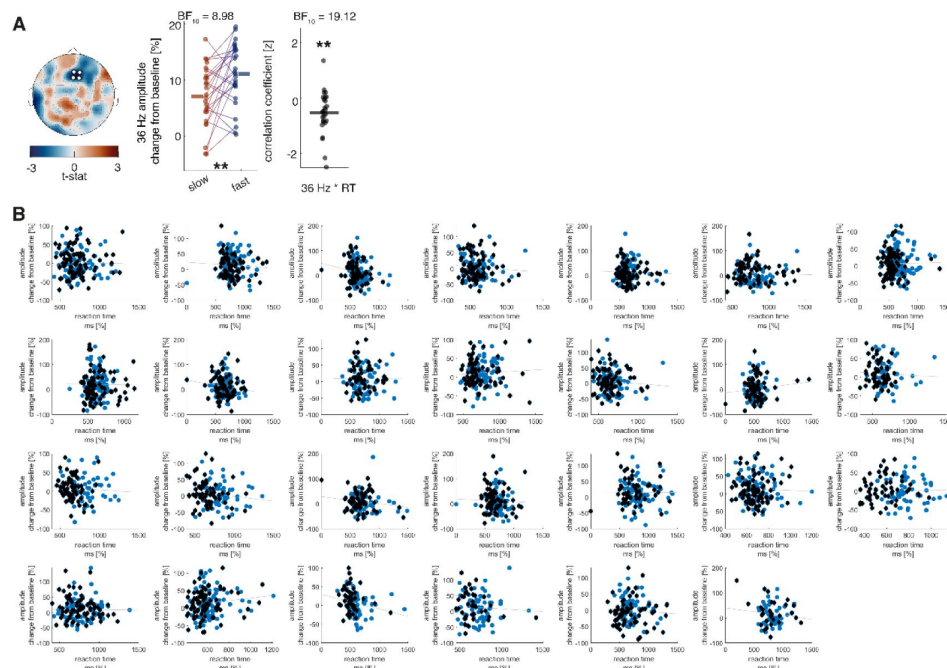


SUPPL Fig. 3

Correlation of prestimulus alpha change from baseline with reaction time in the MEG study.

A, The analysis was performed using a cluster-permutation approach, testing a correlation model against a 0-correlation model. Clusters significantly diverging from the 0-correlation model are presented topographically ($p = .037$). Additionally, median splits between fast and slow reaction time trials ($p = .013$; $t_{(25)} = -2.67$) as well as correlation coefficients ($p = .003$; $t_{(25)} = -3.34$) of these clusters are displayed for all participants. A negative correlation is visible between alpha modulation and reaction times in the last 500 ms before target onset when expecting a visual target. **B**, Correlation between alpha modulation and reaction time for each participant. Black diamonds represent trials from the first block (without distractor) and blue dots represent trials from the second block (with auditory distractor).

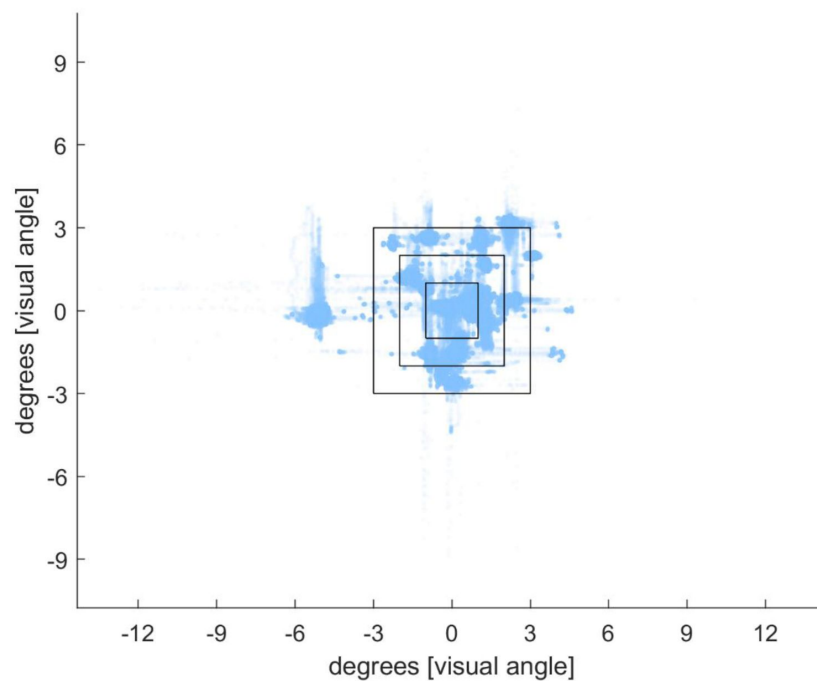




SUPPL Fig. 4

Correlation of 36 Hz change from baseline with reaction time in the MEG study.

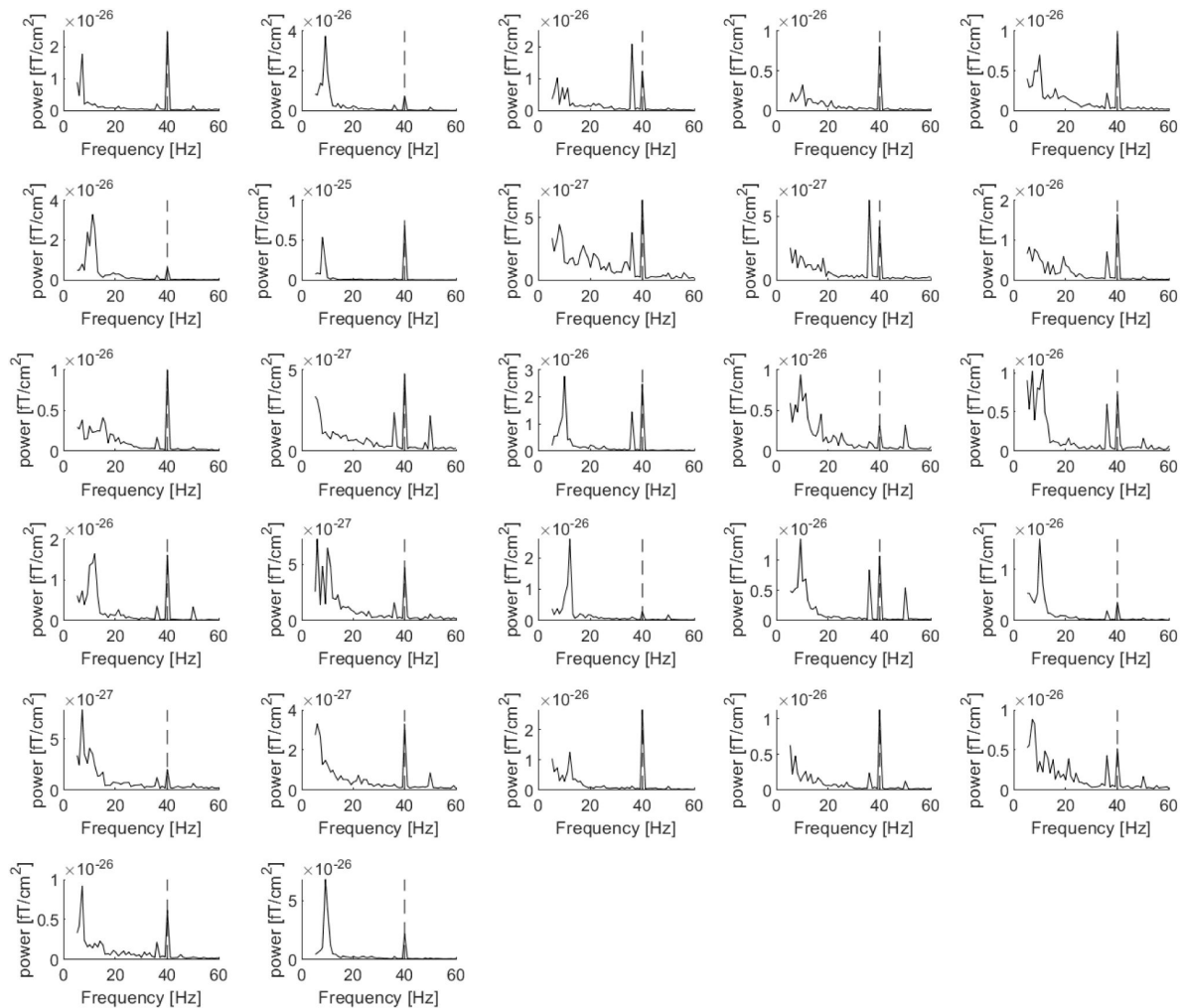
A, The analysis was performed using a cluster-permutation approach, testing a correlation model against a 0-correlation model. Clusters significantly diverging from the 0-correlation model are presented topographically ($p = .040$). Additionally, median splits between fast and slow reaction time trials ($p = .005$; $t_{(25)} = -3.10$) as well as correlation coefficients ($p = .002$; $t_{(25)} = -3.46$) of these clusters are displayed for all participants. A negative correlation is visible between 36 Hz modulation and reaction times in the last 500 ms before target onset when expecting a visual target. **B**, Correlation between 36 Hz modulation and reaction time for each participant. Black diamonds represent trials from the first block (without distractor) and blue dots represent trials from the second block (with auditory distractor).



SUPPL Fig. 5

Illustration of eye-tracking during the cue-to-target interval (2.5 – 0 s before target onset).

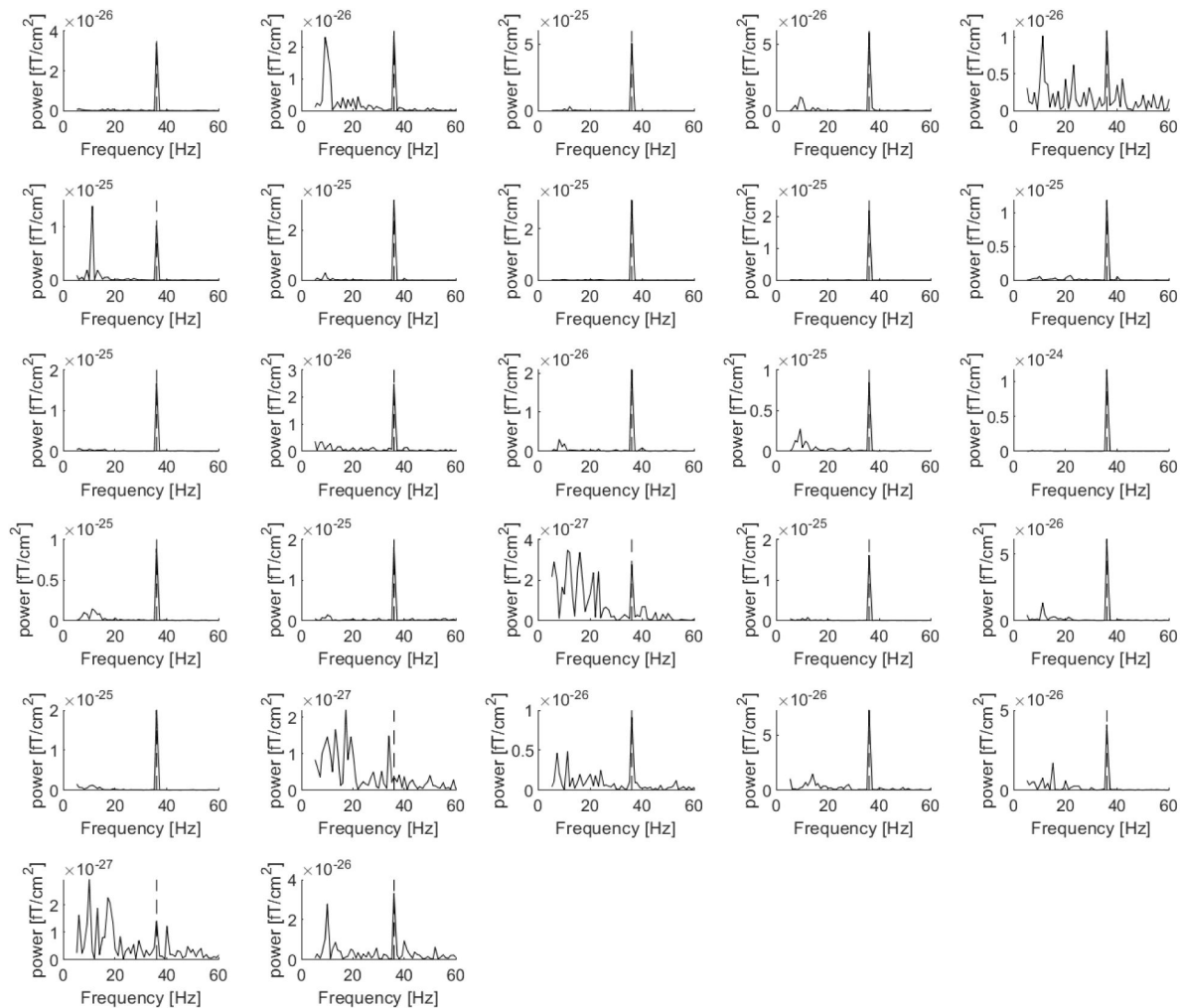
All datapoints of eye-positions during the cue-to-trial interval for all trials and all participants were plotted with 5% visibility on top of each other. Only 3% of datapoints showed eye-movement larger than 3 degrees of visual angle away from the fixation cross.



SUPPL Fig. 6

Individual ERP power spectra of the cue-to-target interval when anticipating an auditory target in the MEG-study.

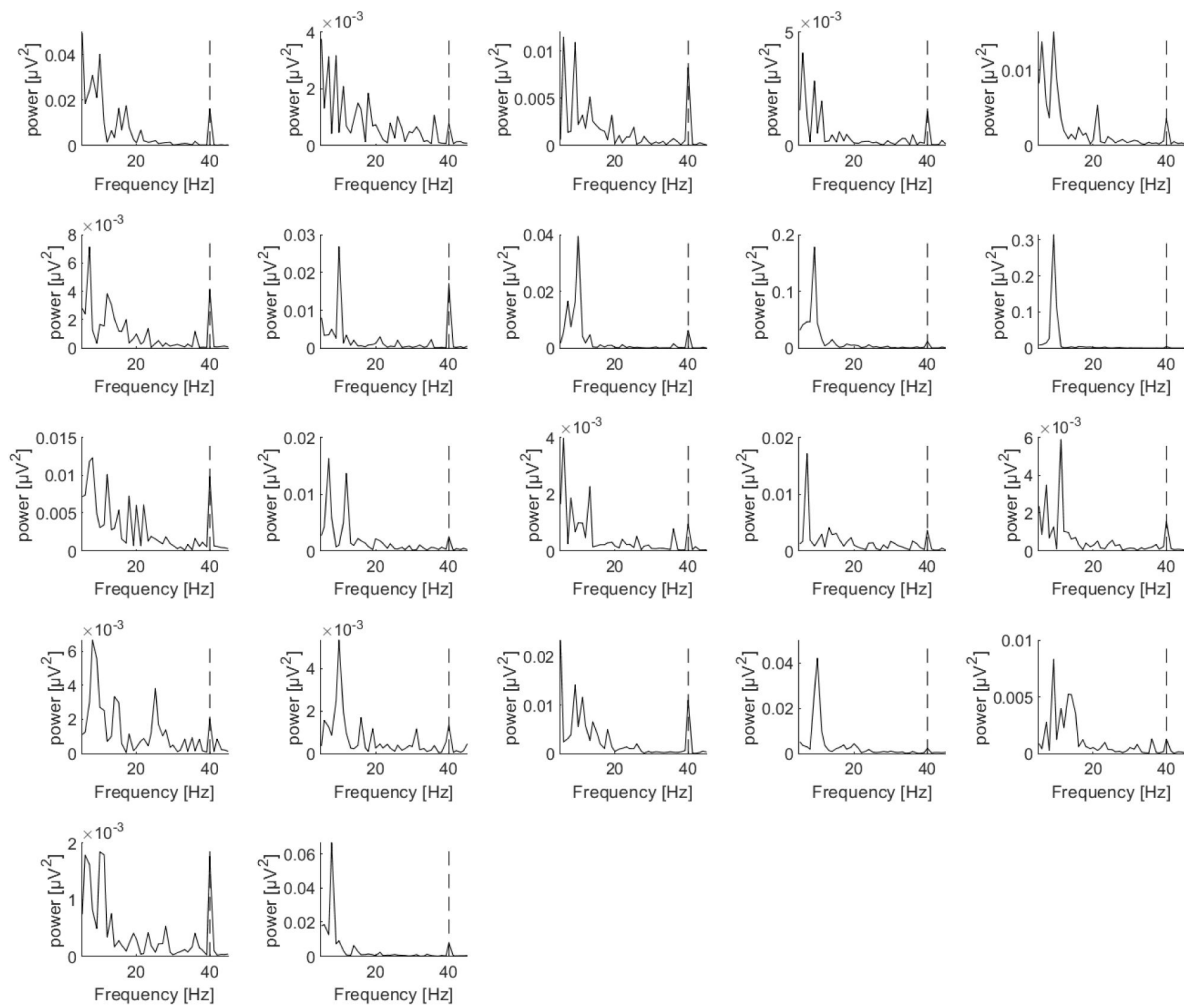
Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 40 Hz (auditory frequency-tagging).



SUPPL Fig. 7

Individual ERP power spectra of the cue-to-target interval when anticipating a visual target in the MEG-study.

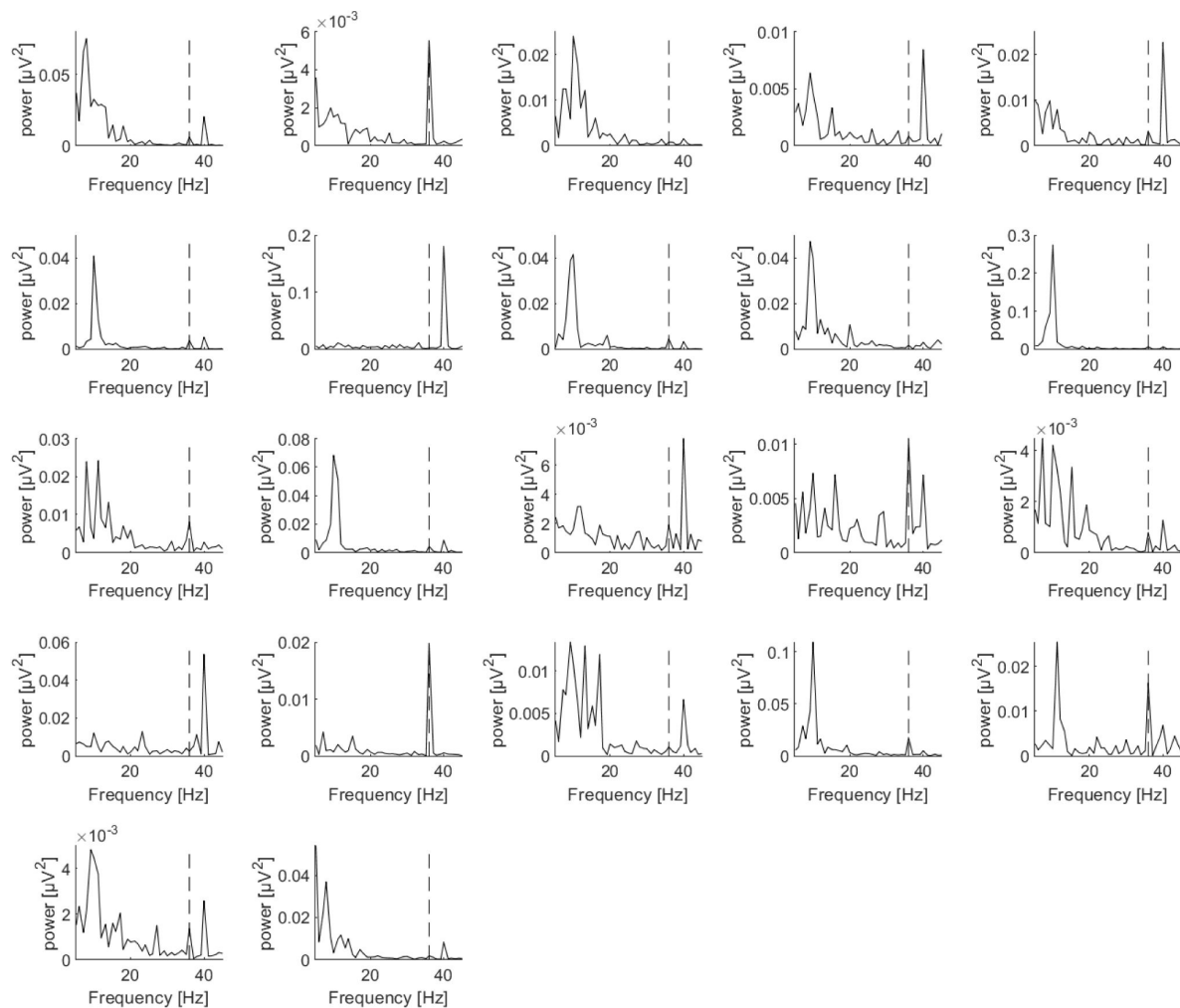
Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 36 Hz (auditory frequency-tagging).



SUPPL Fig. 8

Individual ERP power spectra of the cue-to-target interval when anticipating an auditory target in the EEG-study.

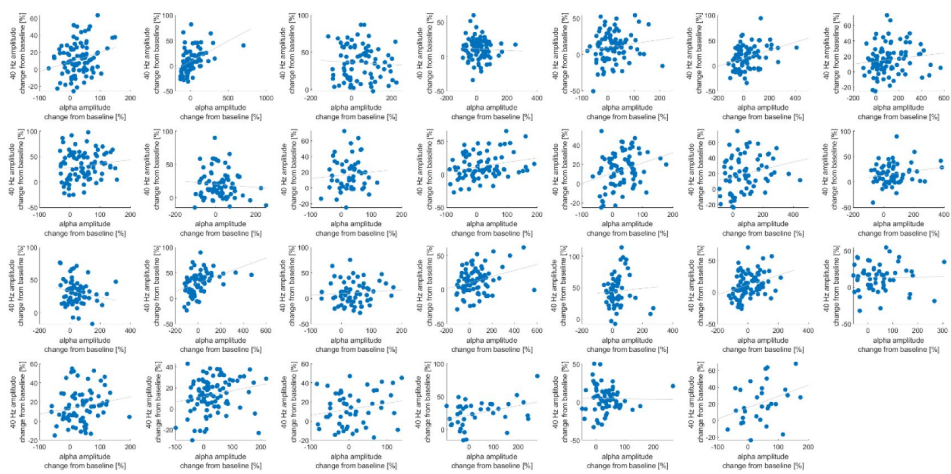
Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 40 Hz (auditory frequency-tagging).



SUPPL Fig. 9

Individual ERP power spectra of the cue-to-target interval when anticipating a visual target in the EEG-study.

Fast-fourier transformation was applied to the averaged trials using a dynamic hanning-tapered sliding time-window of 7 cycles per frequency. The Dotted line represents 36 Hz (auditory frequency-tagging).



SUPPL Fig. 10

Exemplary illustration of the correlation between alpha power (0.5 – 0 s before target onset) and 36 Hz steady-state response (0.5 – 0 s before target onset) for each participant.

Alpha activity was averaged over the significant group difference cluster for alpha condition differences (seed cluster). Frequency-tagging activity was averaged over the significant cluster in the correlation with the alpha seed activity.

Data availability

Codes for analyses and figures will be made available on github and the data will be uploaded to Dryad or OSF

Acknowledgements

This work was made possible by funding support from Facebook Oculus and BBSRC (BB/R018723/1).

References

- Antonov P. A., Chakravarthi R., Andersen S. K 2020) **Too little, too late, and in the wrong place: Alpha band activity does not reflect an active mechanism of selective attention** *NeuroImage* **219**:117006 <https://doi.org/10.1016/j.NEUROIMAGE.2020.117006>
- Bauer A. K. R., Debener S., Nobre A. C 2020) **Synchronisation of Neural Oscillations and Cross-modal Influences** *Trends in Cognitive Sciences* **24**:481–495 <https://doi.org/10.1016/j.tics.2020.03.003>
- Brickwedde M., Bezsudnova Y., Kowalczyk A., Jensen O., Zhigalov A 2022) **Application of rapid invisible frequency tagging for brain computer interfaces** *Journal of Neuroscience Methods* **382**:109726 <https://doi.org/10.1016/j.JNEUMETH.2022.109726>
- Brickwedde M., Schmidt M. D., Krüger M. C., Dinse H. R 2020) **20 Hz Steady-State Response in Somatosensory Cortex During Induction of Tactile Perceptual Learning Through LTP-Like Sensory Stimulation** *Frontiers in Human Neuroscience* <https://doi.org/10.3389/fnhum.2020.00257>
- Colavita F. B. 1974) **Human sensory dominance*** *Perception & Psychophysics* **16**
- Colon E., Legrain V., Mouraux A 2012) **Steady-state evoked potentials to study the processing of tactile and nociceptive somatosensory input in the human brain** *Neurophysiologie Clinique/Clinical Neurophysiology* **42**:315–323 <https://doi.org/10.1016/j.NEUCLI.2012.05.005>
- Crameri F., Shephard G. E., Heron P. J 2020) **The misuse of colour in science communication** *Nature Communications* **11**:1 <https://doi.org/10.1038/s41467-020-19160-7>
- de Jong R., Toffanin P., Harbers M. 2010) **Dynamic crossmodal links revealed by steady-state responses in auditory-visual divided attention** *International Journal of Psychophysiology* **75**:3–15 <https://doi.org/10.1016/j.IJPSYCHO.2009.09.013>
- Dinse H. R., Ragert P., Pleger B., Schwenkreis P., Tegenthoff M 1998) **Pharmacological Modulation of Perceptual Learning and Associated Cortical Reorganization** *Science* **282**:865
- Drijvers L., Jensen O., Spaak E 2021) **Rapid invisible frequency tagging reveals nonlinear integration of auditory and visual information** *Human Brain Mapping* **42**:1138–1152 <https://doi.org/10.1002/hbm.25282>
- Driver J 1996) **Enhancement of selective listening by illusory mislocation of speech sounds due to lip-reading** *Nature* **381**
- Driver J., Spence C 1998) **Cross-modal links in spatial attention** *Philos Trans R Soc Lond B Biol Sci* **353**:1319–1331
- Dugué L., Marque P., VanRullen R 2011) **The phase of ongoing oscillations mediates the causal relation between brain excitation and visual perception** *Journal of Neuroscience* **31**:11889–11893 <https://doi.org/10.1523/JNEUROSCI.1161-11.2011>

- Ergenoglu T., Demiralp T., Bayraktaroglu Z., Ergen M., Beydagi H., Uresin Y 2004) **Alpha rhythm of the EEG modulates visual detection performance in humans** *Cognitive Brain Research* **20**:376–383 <https://doi.org/10.1016/J.COGBRAINRES.2004.03.009>
- Foxe J. J., Simpson G. V, Ahlfors S. P. 1998) **Parieto-occipital ~10Hz activity reflects anticipatory state of visual attention mechanisms** *NeuroReport*
- Foxe J. J., Snyder A. C 2011) **The role of alpha-band brain oscillations as a sensory suppression mechanism during selective attention** *Frontiers in Psychology* **2**:JUL <https://doi.org/10.3389/fpsyg.2011.00154>
- Fu K. M. G., Foxe J. J., Murray M. M., Higgins B. A., Javitt D. C., Schroeder C. E 2001) **Attention-dependent suppression of distracter visual input can be cross-modally cued as indexed by anticipatory parieto-occipital alpha-band oscillations** *Cognitive Brain Research* **12**:145–152 [https://doi.org/10.1016/S0926-6410\(01\)00034-9](https://doi.org/10.1016/S0926-6410(01)00034-9)
- Gundlach C., Moratti S., Forschack N., Müller M. M 2020) **Spatial Attentional Selection Modulates Early Visual Stimulus Processing Independently of Visual Alpha Modulations** *Cerebral Cortex* **30**:3686–3703 <https://doi.org/10.1093/cercor/bhz335>
- Gutteling T. P., Sillekens L., Lavie N., Jensen O 2022) **Alpha oscillations reflect suppression of distractors with increased perceptual load** *Progress in Neurobiology* **214** <https://doi.org/10.1016/J.PNEUROBIO.2022.102285>
- Haegens S., Barczak A., Musacchia G., Lipton M. L., Mehta A. D., Lakatos P., Schroeder C. E 2015) **Laminar profile and physiology of the α rhythm in primary visual, auditory, and somatosensory regions of neocortex** *Journal of Neuroscience* **35**:14341–14352 <https://doi.org/10.1523/JNEUROSCI.0600-15.2015>
- Haegens S., Nácher V., Luna R., Romo R., Jensen O 2011) **α -Oscillations in the monkey sensorimotor network influence discrimination performance by rhythmical inhibition of neuronal spiking** *Proceedings of the National Academy of Sciences* **108**:19377–19382 <https://doi.org/10.1073/PNAS.1117190108>
- Hari R., Hämäläinen M., Joutsiniemi S. L 1989) **Neuromagnetic steady-state responses to auditory stimuli** *Journal of the Acoustical Society of America* **86**:1033–1039 <https://doi.org/10.1121/1.398093>
- Holmes C. J., Hoge R., Collins L., Woods R., Toga A. W., Evans A. C 1998) **Enhancement of MR images using registration for signal averaging** *Journal of Computer Assisted Tomography* **22**:324–333 <https://doi.org/10.1097/00004728-199803000-00032>
- Jacoby O., Hall S. E., Mattingley J. B 2012) **A crossmodal crossover: Opposite effects of visual and auditory perceptual load on steady-state evoked potentials to irrelevant visual stimuli** *NeuroImage* **61**:1050–1058 <https://doi.org/10.1016/J.NEUROIMAGE.2012.03.040>
- Jensen O., Mazaheri A 2010) **Shaping functional architecture by oscillatory alpha activity: Gating by inhibition** *Frontiers in Human Neuroscience* **4** <https://doi.org/10.3389/fnhum.2010.00186>

- Kelly S. P., Lalor E. C., Reilly R. B., Foxe J. J 2006) **Increases in alpha oscillatory power reflect an active retinotopic mechanism for distracter suppression during sustained visuospatial attention** *Journal of Neurophysiology* **95**:3844–3851 <https://doi.org/10.1152/jn.01234.2005>
- Klatt L.-I., Getzmann S., Schneider D 2021) **Attentional Modulations of Alpha Power Are Sensitive to the Task-relevance of Auditory Spatial Information** *BioRxiv* <https://doi.org/10.1101/2021.02.12.430942>
- Klimesch W., Sauseng P., Hanslmayr S 2007) **EEG alpha oscillations: The inhibition-timing hypothesis** *Brain Research Reviews* **53**:63–88 <https://doi.org/10.1016/J.BRAINRESREV.2006.06.003>
- Lakatos P., Karmos G., Mehta A. D., Ulbert I., Schroeder C. E 2008) **Entrainment of neuronal oscillations as a mechanism of attentional selection** *Science (New York, N.Y.)* **320**:110–113 <https://doi.org/10.1126/SCIENCE.1154735>
- Maris E., Oostenveld R 2007) **Nonparametric statistical testing of EEG- and MEG-data** *Journal of Neuroscience Methods* **164**:177–190 <https://doi.org/10.1016/J.JNEUMETH.2007.03.024>
- Marzoll A., Saygi T., Dinse H. R 2018) **The effect of LTP- and LTD-like visual stimulation on modulation of human orientation discrimination** *Scientific Reports* **8**:1 <https://doi.org/10.1038/s41598-018-34276-z>
- Mazaheri A., van Schouwenburg M. R., Dimitrijevic A., Denys D., Cools R., Jensen O. 2014) **Region-specific modulations in oscillatory alpha activity serve to facilitate processing in the visual and auditory modalities** *NeuroImage* **87**:356–362 <https://doi.org/10.1016/J.NEUROIMAGE.2013.10.052>
- Müller M. M., Hillyard S 2000) **Concurrent recording of steady-state and transient event-related potentials as indices of visual-spatial selective attention** *Clinical Neurophysiology* **111**:1544–1552 [https://doi.org/10.1016/S1388-2457\(00\)00371-0](https://doi.org/10.1016/S1388-2457(00)00371-0)
- Müller M. M., Picton T. W., Valdes-Sosa P., Riera J., Teder-Sälejärvi W. A., Hillyard S. A 1998) **Effects of spatial selective attention on the steady-state visual evoked potential in the 20– 28 Hz range** *Cognitive Brain Research* **6**:249–261 [https://doi.org/10.1016/S0926-6410\(97\)00036-0](https://doi.org/10.1016/S0926-6410(97)00036-0)
- Okazaki Y. O., De Weerd P., Haegens S., Jensen O. 2014) **Hemispheric lateralization of posterior alpha reduces distracter interference during face matching** *Brain Research* **1590**:56–64 <https://doi.org/10.1016/J.BRAINRES.2014.09.058>
- Pantev C., Roberts L. E., Elbert T., Roß B., Wienbruch C 1996) **Tonotopic organization of the sources of human auditory steady-state responses** *Hearing Research* **101**:62–74 [https://doi.org/10.1016/S0378-5955\(96\)00133-5](https://doi.org/10.1016/S0378-5955(96)00133-5)
- Popov T., Gips B., Kastner S., Jensen O 2019) **Spatial specificity of alpha oscillations in the human visual system** *Human Brain Mapping* **40**:4432–4440 <https://doi.org/10.1002/hbm.24712>
- Popov T., Gips B., Weisz N., Jensen O 2021) **Brain areas associated with visual spatial attention display topographic organization during auditory spatial attention** *BioRxiv* <https://doi.org/10.1101/2021.03.15.435371>

- Porcu E., Keitel C., Müller M. M 2013) **Concurrent visual and tactile steady-state evoked potentials index allocation of inter-modal attention: A frequency-tagging study** *Neuroscience Letters* **556**:113–117 <https://doi.org/10.1016/j.NEULET.2013.09.068>
- Regan D 1982) **Comparison of transient and steady-state methods** *Ann N Y Acad Sci*
- Saupe K., Schröger E., Andersen S. K., Müller M. M 2009) **Neural mechanisms of intermodal sustained selective attention with concurrently presented auditory and visual stimuli** *Frontiers in Human Neuroscience* **3** <https://doi.org/10.3389/neuro.09.058.2009>
- Sekihara K., Nagarajan S. S., Poeppel D., Marantz A 2004) **Asymptotic SNR of scalar and vector minimum-variance beamformers for neuromagnetic source reconstruction** *IEEE Transactions on Biomedical Engineering* **51**:1726–1734 <https://doi.org/10.1109/TBME.2004.827926>
- Sinnett S., Soto-Faraco S., Spence C 2008) **The co-occurrence of multisensory competition and facilitation** *Acta Psychologica* **128**:153–161 <https://doi.org/10.1016/j.ACTPSY.2007.12.002>
- Snyder A 1992) **Steady-state vibration evoked potentials: description of technique and characterization of responses** *Electroencephalography and Clinical Neurophysiology/Evoked Potentials Section* **84**:257–268 [https://doi.org/10.1016/0168-5597\(92\)90007-X](https://doi.org/10.1016/0168-5597(92)90007-X)
- Spaak E., Bonnefond M., Maier A., Leopold D. A., Jensen O 2012) **Layer-Specific Entrainment of Gamma-Band Neural Activity by the Alpha Rhythm in Monkey Visual Cortex** *Current Biology* **22**:2313–2318 <https://doi.org/10.1016/j.CUB.2012.10.020>
- Spence C 2009) **Explaining the Colavita visual dominance effect** *Progress in Brain Research* **176**:245–258 [https://doi.org/10.1016/S0079-6123\(09\)17615-X](https://doi.org/10.1016/S0079-6123(09)17615-X)
- Spence C., Driver J 1996) **Audiovisual Links in Endogenous Covert Spatial Attention** *Journal of Experimental Psychology: Human Perception and Performance* **22**:1005–1030
- Stapells D. R., Linden D., Braxton Suffield J., Hamel G., Picton T. W., Wp M. T. 1984) **Human auditory steady state potentials** *Ear Hear* **5**
- Thut G., Nietzel A., Brandt S. A., Pascual-Leone A 2006) **Alpha-band electroencephalographic activity over occipital cortex indexes visuospatial attention bias and predicts visual target detection** *J Neurosci* <https://doi.org/10.1523/JNEUROSCI.0875-06.2006>
- Tobimatsu S., Zhang Y. M., Kato M 1999) **Steady-state vibration somatosensory evoked potentials: physiological characteristics and tuning function** *Clinical Neurophysiology* **110**:1953–1958 [https://doi.org/10.1016/S1388-2457\(99\)00146-7](https://doi.org/10.1016/S1388-2457(99)00146-7)
- Toffanin P., de Jong R., Johnson A., Martens S. 2009) **Using frequency tagging to quantify attentional deployment in a visual divided attention task** *International Journal of Psychophysiology* **72**:289–298 <https://doi.org/10.1016/j.IJPSYCHO.2009.01.006>
- Tomassini A., Ambrogioni L., Medendorp W. P., Maris E 2017) **Theta oscillations locked to intended actions rhythmically modulate perception** *eLife* **6** <https://doi.org/10.7554/eLife.25618.001>
- Tzourio-Mazoyer N., Landeau B., Papathanassiou D., Crivello F., Etard O., Delcroix N., Mazoyer B., Joliot M 2002) **Automated anatomical labeling of activations in SPM using a**

macroscopic anatomical parcellation of the MNI MRI single-subject brain *NeuroImage* **15**:273–289 <https://doi.org/10.1006/nimg.2001.0978>

Uemura J.-I., Hoshino A., Igarashi Go, Matsui Y., Chishima M., Hoshiyama, | Minoru, Mazaheri A. 2021) **Pre-stimulus alpha oscillation and post-stimulus cortical activity differ in localization between consciously perceived and missed near-threshold somatosensory stimuli** *Eur J Neurosci* <https://doi.org/10.1111/ejn.15388>

Van Diepen R. M., Cohen M. X., Denys D., Mazaheri A. 2015) **Attention and temporal expectations modulate power, not phase, of ongoing alpha oscillations** *Journal of Cognitive Neuroscience* **27**:1573–1586 https://doi.org/10.1162/jocn_a_00803

Van Diepen R. M., Foxe J. J., Mazaheri A. 2019) **The functional role of alpha-band activity in attentional processing: the current zeitgeist and future outlook** *Current Opinion in Psychology* **29**:229–238 <https://doi.org/10.1016/j.COPSYC.2019.03.015>

van Diepen R. M., Mazaheri A. 2017) **Cross-sensory modulation of alpha oscillatory activity: suppression, idling, and default resource allocation** *European Journal of Neuroscience* **45**:1431–1438 <https://doi.org/10.1111/ejn.13570>

Van Dijk H., Schoffelen J. M., Oostenveld R., Jensen O. 2008) **Prestimulus oscillatory activity in the alpha band predicts visual discrimination ability** *Journal of Neuroscience* **28**:1816–1823 <https://doi.org/10.1523/JNEUROSCI.1853-07.2008>

Worden M. S., Foxe J. J., Wang N., Simpson G. V. 2000) **Anticipatory biasing of visuospatial attention indexed by retinotopically specific alpha-band electroencephalography increases over occipital cortex** *J Neurosci* <https://doi.org/10.1523/JNEUROSCI.20-06-j0002.2000>

Yang X., Fiebelkorn I. C., Jensen O., Knight R. T., Kastner S 2023) **Differential neural mechanisms underlie cortical gating of visual spatial attention mediated by alpha-band oscillations** *BioRxiv* <https://doi.org/10.1101/2023.08.21.553303>

Zhigalov A., Herring J. D., Herpers J., Bergmann T. O., Jensen O 2019) **Probing cortical excitability using rapid frequency tagging** *NeuroImage* **195**:59–66 <https://doi.org/10.1016/j.NEUROIMAGE.2019.03.056>

Zhigalov A., Jensen O 2020) **Alpha oscillations do not implement gain control in early visual cortex but rather gating in parieto-occipital regions** *Human Brain Mapping* **41**:5176–5186 <https://doi.org/10.1002/hbm.25183>

Zumer J. M., Scheeringa R., Schoffelen J. M., Norris D. G., Jensen O 2014) **Occipital Alpha Activity during Stimulus Processing Gates the Information Flow to Object-Selective Cortex** *PLoS Biology* **12**:10 <https://doi.org/10.1371/journal.pbio.1001965>

Author information

Marion Brickwedde

Department of Child and Adolescent Psychiatry, Charité - Universitätsmedizin Berlin, Berlin, Germany, Physikalisch Technische Bundesanstalt, Berlin, Germany, Centre for Human Brain Health, School of Psychology, University of Birmingham, Birmingham, United Kingdom

ORCID iD: [0000-0002-3461-038X](https://orcid.org/0000-0002-3461-038X)

For correspondence: marion.brickwedde@outlook.com

Rupali Limachya

Centre for Human Brain Health, School of Psychology, University of Birmingham,
Birmingham, United Kingdom

Roksana Markiewicz

Istituto Italiano di Tecnologia, Genova, Italy
ORCID iD: [0000-0002-5311-8008](https://orcid.org/0000-0002-5311-8008)

Emma Sutton

Centre for Human Brain Health, School of Psychology, University of Birmingham,
Birmingham, United Kingdom

Christopher Postzich

Max-Planck Research Institute, Leipzig, Germany

Kimron Shapiro

Centre for Human Brain Health, School of Psychology, University of Birmingham,
Birmingham, United Kingdom
ORCID iD: [0000-0002-0419-1460](https://orcid.org/0000-0002-0419-1460)

Ole Jensen

Centre for Human Brain Health, School of Psychology, University of Birmingham,
Birmingham, United Kingdom
ORCID iD: [0000-0001-8193-8348](https://orcid.org/0000-0001-8193-8348)

Ali Mazaheri

Centre for Human Brain Health, School of Psychology, University of Birmingham,
Birmingham, United Kingdom
ORCID iD: [0000-0001-5732-7555](https://orcid.org/0000-0001-5732-7555)

Editors

Reviewing Editor

Iris Groen

University of Amsterdam, Amsterdam, Netherlands

Senior Editor

Huan Luo

Peking University, Beijing, China

Reviewer #1 (Public review):

In this paper by Brickwedde et al., the authors observe an increase in posterior alpha when anticipating auditory as opposed to visual targets. The authors also observe an enhancement in both visual and auditory steady-state sensory evoked potentials in anticipation of auditory targets, in correlation with enhanced occipital alpha. The authors conclude that alpha does not reflect inhibition of early sensory processing, but rather orchestrates signal transmission to later stages of the sensory processing stream. However, there are several major concerns that need to be addressed in order to draw this conclusion.

First, I am not convinced that the frequency tagging method and the associated analyses are adequate for dissociating visual vs auditory steady-state sensory evoked potentials.

Second, if the authors want to propose a general revision for the function of alpha, it would be important to show that alpha effects in the visual cortex for visual perception are analogous to alpha effects in the auditory cortex for auditory perception.

Third, the authors propose an alternative function for alpha - that alpha orchestrates signal transmission to later stages of the sensory processing stream. However, the supporting evidence for this alternative function is lacking. I will elaborate on these major concerns below.

(1) Potential bleed-over across frequencies in the spectral domain is a major concern for all of the results in this paper. The fact that alpha power, 36Hz and 40Hz frequency-tagged amplitude and 4Hz intermodulation frequency power is generally correlated with one another amplifies this concern. The authors are attaching specific meaning to each of these frequencies, but perhaps there is simply a broadband increase in neural activity when anticipating an auditory target compared to a visual target?

(2) Moreover, 36Hz visual and 40Hz auditory signals are expected to be filtered in the neocortex. Applying standard filters and Hilbert transform to estimate sensory evoked potentials appears to rely on huge assumptions that are not fully substantiated in this paper. In Figure 4, 36Hz "visual" and 40Hz "auditory" signals seem largely indistinguishable from one another, suggesting that the analysis failed to fully demix these signals.

(3) The asymmetric results in the visual and auditory modalities preclude a modality-general conclusion about the function of alpha. However, much of the language seems to generalize across sensory modalities (e.g., use of the term 'sensory' rather than 'visual').

(4) In this vein, some of the conclusions would be far more convincing if there was at least a trend towards symmetry in source-localized analyses of MEG signals. For example, how does alpha power in the primary auditory cortex (A1) compare when anticipating auditory vs visual target? What do the frequency-tagged visual and auditory responses look like when just looking at the primary visual cortex (V1) or A1?

(5) Blinking would have a huge impact on the subject's ability to ignore the visual distractor. The best thing to do would be to exclude from analysis all trials where the subjects blinked during the cue-to-target interval. The authors mention that in the MEG experiment, "To remove blinks, trials with very large eye-movements (> 10 degrees of visual angle) were removed from the data (See supplement Fig. 5)." This sentence needs to be clarified since eye-movements cannot be measured during blinking. In addition, it seems possible to remove putative blink trials from EEG experiments as well, since blinks can be detected in the EEG signals.

(6) It would be interesting to examine the neutral cue trials in this task. For example, comparing auditory vs visual vs neutral cue conditions would be indicative of whether alpha was actively recruited or actively suppressed. In addition, comparing spectral activity during cue-to-target period on neutral-cue auditory correct vs incorrect trials should mimic the comparison of auditory-cue vs visual-cue trials. Likewise, neutral-cue visual correct vs incorrect trials should mimic the attention-related differences in visual-cue vs auditory-cue trials.

(7) In the abstract, the authors state that "This implies that alpha modulation does not solely regulate 'gain control' in early sensory areas but rather orchestrates signal transmission to later stages of the processing stream." However, I don't see any supporting evidence for the latter claim, that alpha orchestrates signal transmission to later stages of the processing

stream. If the authors are claiming an alternative function to alpha, this claim should be strongly substantiated.

<https://doi.org/10.7554/eLife.106050.1.sa3>

Reviewer #2 (Public review):

Brickwedde et al. investigate the role of alpha oscillations in allocating intermodal attention. A first EEG study is followed up with a MEG study that largely replicates the pattern of results (with small to be expected differences). They conclude that a brief increase in the amplitude of auditory and visual stimulus-driven continuous (steady-state) brain responses prior to the presentation of an auditory - but not visual - target speaks to the modulating role of alpha that leads them to revise a prevalent model of gating-by-inhibition.

Overall, this is an interesting study on a timely question, conducted with methods and analysis that are state-of-the-art. I am particularly impressed by the author's decision to replicate the earlier EEG experiment in MEG following the reviewer's comments on the original submission. Evidently, great care was taken to accommodate the reviewer's suggestions.

Nevertheless, I am struggling with the report for two main reasons: It is difficult to follow the rationale of the study, due to structural issues with the narrative and missing information or justifications for design and analysis decisions, and I am not convinced that the evidence is strong, or even relevant enough for revising the mentioned alpha inhibition theory. Both points are detailed further below.

Strength/relevance of evidence for model revision: The main argument rests on 1) a rather sustained alpha effect following the modality cue, 2) a rather transient effect on steady-state responses just before the expected presentation of a stimulus, and 3) a correlation between those two. Wouldn't the authors expect a sustained effect on sensory processing, as measured by steady-state amplitude irrespective of which of the scenarios described in Figure 1A (original vs revised alpha inhibition theory) applies? Also, doesn't this speak to the role of expectation effects due to consistent stimulus timing? An alternative explanation for the results may look like this: Modality-general increased steady-state responses prior to the expected audio stimulus onset are due to increased attention/vigilance. This effect may be exclusive (or more pronounced) in the attend-audio condition due to higher precision in temporal processing in the auditory sense or, vice versa, too smeared in time due to the inferior temporal resolution of visual processing for the attend-vision condition to be picked up consistently. As expectation effects will build up over the course of the experiment, i.e., while the participant is learning about the consistent stimulus timing, the correlation with alpha power may then be explained by a similar but potentially unrelated increase in alpha power over time.

Structural issues with the narrative and missing information: Here, I am mostly concerned with how this makes the research difficult to access for the reader. I list the major points below:

In the introduction the authors pit the original idea about alpha's role in gating against some recent contradictory results. If it's the aim of the study to provide evidence for either/or, predictions for the results from each perspective are missing. Also, it remains unclear how this relates to the distinction between original vs revised alpha inhibition theory (Fig. 1A). Relatedly if this revision is an outcome rather than a postulation for this study, it shouldn't be featured in the first figure.

The analysis of the intermodulation frequency makes a surprise entrance at the end of the Results section without an introduction as to its relevance for the study. This is provided only in the discussion, but with reference to multisensory integration, whereas the main focus of the study is focussed attention on one sense. (Relatedly, the reference to "theta oscillations" in this sections seems unclear without a reference to the overlapping frequency range, and potentially more explanation.) Overall, if there's no immediate relevance to this analysis, I would suggest removing it.

<https://doi.org/10.7554/eLife.106050.1.sa2>

Reviewer #3 (Public review):

Brickwedde et al. attempt to clarify the role of alpha in sensory gain modulation by exploring the relationship between attention-related changes in alpha and attention-related changes in sensory-evoked responses, which surprisingly few studies have examined given the prevalence of the alpha inhibition hypothesis. The authors use robust methods and provide novel evidence that alpha likely exhibits inhibitory control over later processing, as opposed to early sensory processing, by providing source-localization data in a cross-modal attention task.

This paper seems very strong, particularly given that the follow-up MEG study both (a) clarifies the task design and separates the effect of distractor stimuli into other experimental blocks, and (b) provides source-localization data to more concretely address whether alpha inhibition is occurring at or after the level of sensory processing, and (c) replicates most of the EEG study's key findings.

There are some points that would be helpful to address to bolster the paper. First, the introduction would benefit from a somewhat deeper review of the literature, not just reviewing when the effects of alpha seem to occur, but also addressing how the effect can change depending on task and stimulus design (see review by Morrow, Elias & Samaha (2023). Additionally, the discussion could benefit from more cautionary language around the revision of the alpha inhibition account. For example, it would be helpful to address some of the possible discrepancies between alpha and SSEP measures in terms of temporal specificity, SNR, etc. (see Peylo, Hilla, & Sauseng, 2021). The authors do a good job speculating as to why they found differing results from previous cross-modal attention studies, but I'm also curious whether the authors think that alpha inhibition/modulation of sensory signals would have been different had the distractors been within the same modality or whether the cues indicated target location, rather than just modality, as has been the case in so much prior work?

Overall, the analyses and discussion are quite comprehensive, and I believe this paper to be an excellent contribution to the alpha-inhibition literature.

<https://doi.org/10.7554/eLife.106050.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

In this paper by Brickwedde et al., the authors observe an increase in posterior alpha when anticipating auditory as opposed to visual targets. The authors also observe an enhancement in both visual and auditory steady-state sensory evoked potentials in

anticipation of auditory targets, in correlation with enhanced occipital alpha. The authors conclude that alpha does not reflect inhibition of early sensory processing, but rather orchestrates signal transmission to later stages of the sensory processing stream. However, there are several major concerns that need to be addressed in order to draw this conclusion.

First, I am not convinced that the frequency tagging method and the associated analyses are adequate for dissociating visual vs auditory steady-state sensory evoked potentials.

Second, if the authors want to propose a general revision for the function of alpha, it would be important to show that alpha effects in the visual cortex for visual perception are analogous to alpha effects in the auditory cortex for auditory perception.

Third, the authors propose an alternative function for alpha - that alpha orchestrates signal transmission to later stages of the sensory processing stream. However, the supporting evidence for this alternative function is lacking. I will elaborate on these major concerns below.

(1) Potential bleed-over across frequencies in the spectral domain is a major concern for all of the results in this paper. The fact that alpha power, 36Hz and 40Hz frequency-tagged amplitude and 4Hz intermodulation frequency power is generally correlated with one another amplifies this concern. The authors are attaching specific meaning to each of these frequencies, but perhaps there is simply a broadband increase in neural activity when anticipating an auditory target compared to a visual target?

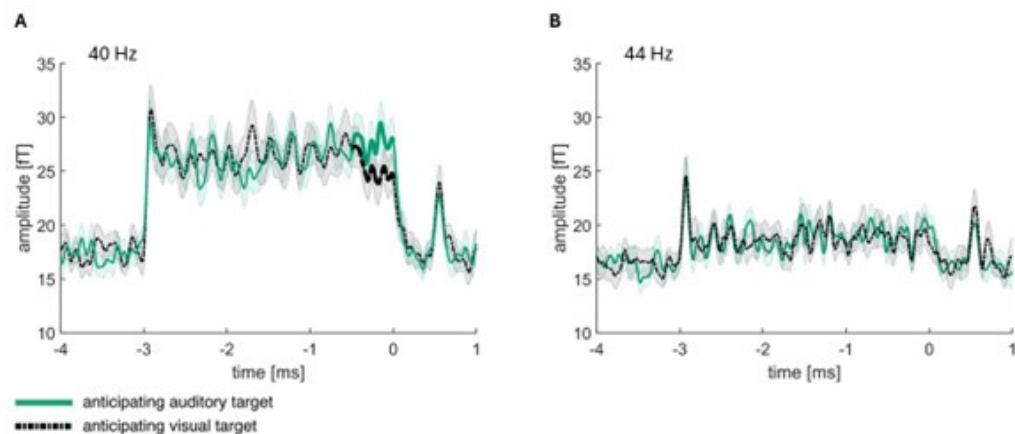
We appreciate the reviewer's insightful comment regarding the potential bleed-over across frequencies in the spectral domain. We fully acknowledge that the trade-off between temporal and frequency resolution is a challenge, particularly given the proximity of the frequencies we are examining.

To address this concern, we performed additional analyses to investigate whether there is indeed a broadband increase in neural activity when anticipating an auditory target as compared to a visual target, as opposed to distinct frequency-specific effects. Our results show that the bleed-over between frequencies is minimal and does not significantly affect our findings. Specifically, we repeated the analyses using the same filter and processing steps for the 44 Hz frequency. At this frequency, we did not observe any significant differences between conditions.

These findings suggest that the effects we report are indeed specific to the 40 Hz frequency band and not due to a general broadband increase in neural activity. We hope this addresses the reviewer's concern and strengthens the validity of our frequency-specific results.

Author response image 1.

Illustration of bleeding over effects over a span of 4 Hz. A, 40 Hz frequency-tagging data over the significant cluster differing between when expecting an auditory versus a visual target (identical to Fig. 9 in the manuscript). B, 44 Hz signal over the same cluster chosen for A. The analysis was identical with the analysis performed in A, apart from the frequency for the band-pass filter.



We do, however, not specifically argue against the possibility of a broadband increase when anticipating an auditory compared to a visual target. But even a broadband-increase would directly contradict the alpha inhibition hypothesis, which poses that an increase in alpha completely disengages the whole cortex. We will clarify this point in the revised manuscript.

(2) Moreover, 36Hz visual and 40Hz auditory signals are expected to be filtered in the neocortex. Applying standard filters and Hilbert transform to estimate sensory evoked potentials appears to rely on huge assumptions that are not fully substantiated in this paper. In Figure 4, 36Hz "visual" and 40Hz "auditory" signals seem largely indistinguishable from one another, suggesting that the analysis failed to fully demix these signals.

We appreciate the reviewer's insightful concern regarding the filtering and demixing of the 36 Hz visual and 40 Hz auditory signals, and we share the same reservations about the reliance on standard filters and the Hilbert transform method.

To address this, we would like to draw attention to Author response image 1, which demonstrates that a 4 Hz difference is sufficient to effectively demix the signals using our chosen filtering and Hilbert transform approach. We believe that the reason the 36 Hz visual and 40 Hz auditory signals show similar topographies lies not in incomplete demixing but rather in the possibility that this condition difference reflects sensory integration, rather than signal contamination.

This interpretation is further supported by our findings with the intermodulation frequency at 4 Hz, which also suggests cross-modal integration. Furthermore, source localization analysis revealed that the strongest condition differences were observed in the precuneus, an area frequently associated with sensory integration processes. We will expand on this in the discussion section to better clarify this point.

(3) The asymmetric results in the visual and auditory modalities preclude a modality-general conclusion about the function of alpha. However, much of the language seems to generalize across sensory modalities (e.g., use of the term 'sensory' rather than 'visual').

We thank the reviewer for pointing this out and agree that in some cases we have not made a good enough distinction between visual and sensory. We will make sure, that when using 'sensory', we either describe overall theories, which are not visual-exclusive or refer to the possibility of a broad sensory increase. However, when directly discussing our results and the interpretation thereof, we will now use 'visual' in the revised manuscript.

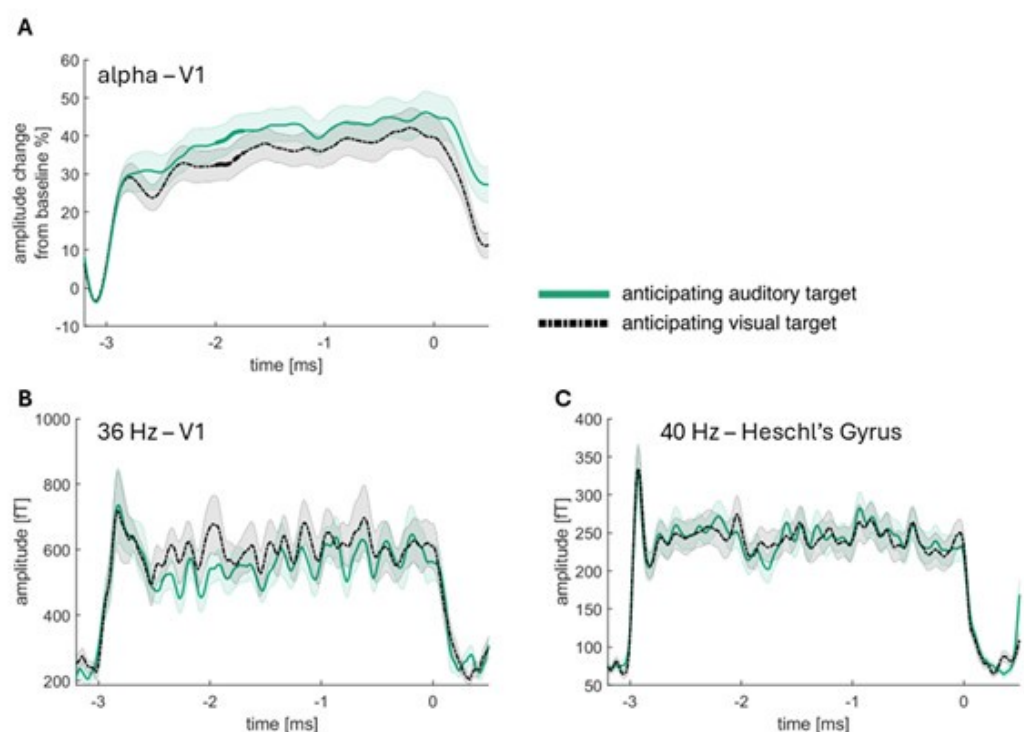
(4) In this vein, some of the conclusions would be far more convincing if there was at least a trend towards symmetry in source-localized analyses of MEG signals. For example, how does alpha power in the primary auditory cortex (A1) compare when anticipating auditory vs visual target? What do the frequency-tagged visual and auditory responses look like when just looking at the primary visual cortex (V1) or A1?

We thank the reviewer for this important suggestion and have added a virtual channel analysis. We were however, not interested in alpha power in primary auditory cortex, as we were specifically interested in the posterior alpha, which is usually increased when expecting an auditory compared to a visual target (and used to be interpreted as a blanket inhibition of the visual cortex). We will improve upon the clarity concerning this point in the manuscript.

We have however, followed the reviewer's suggestion of a virtual channel analysis, showing that the condition differences are not observable in primary visual cortex for the 36 Hz visual signal and in primary auditory cortex for the 40 Hz auditory signal. Our data clearly shows that there is an alpha condition difference in V1, while there no condition difference for 36 Hz in V1 and for 40 Hz in Heschl's Gyrus (see Author response image 2).

Author response image 2.

Virtual channels for V1 and Heschl's gyrus. A, alpha power for the virtual channel created in V1 (Calcerine_L and Calcerine_R from AAL atlas; Tzourio-Mazoyer et al., 2002, NeuroImage). A cluster permutation analysis over time (between -2 and 0) revealed a significant condition difference between ~ -2 and -1.7 s ($p = 0.0449$). B, 36 Hz frequency-tagging signal for the virtual channel created in V1 (equivalent to the procedure in A). The same cluster permutation as performed in A revealed no significant condition differences. C, 40 Hz frequency-tagging signal for the virtual channel created in Heschl's gyrus (Heschl_L and Heschl_R from AAL atlas; Tzourio-Mazoyer et al., 2002, NeuroImage). The same cluster permutation as performed in A revealed no significant condition differences.



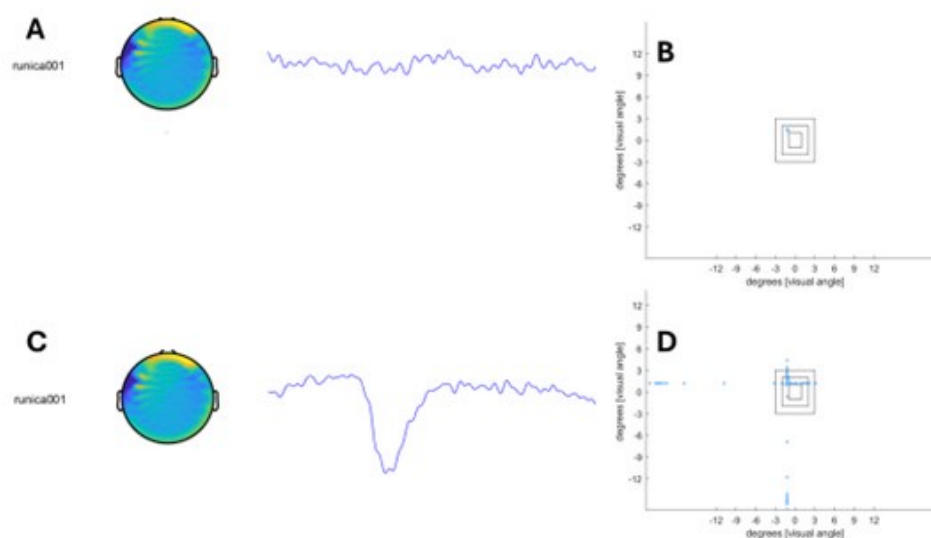
(5) Blinking would have a huge impact on the subject's ability to ignore the visual distractor. The best thing to do would be to exclude from analysis all trials where the subjects blinked during the cue-to-target interval. The authors mention that in the MEG experiment, "To remove blinks, trials with very large eye-movements (> 10 degrees of visual angle) were removed from the data (See supplement Fig. 5)." This sentence needs to be clarified since eye-movements cannot be measured during blinking. In addition, it seems possible to remove putative blink trials from EEG experiments as well, since blinks can be detected in the EEG signals.

We thank the reviewer for mentioning that we were making this point confusing. From the MEG-data, we removed eyeblinks using ICA. Alone for the supplementary Fig. 5 analysis, we used the eye-tracking data to confirm that participants were in fact fixating the centre of the screen. For this analysis, we removed trials with blinks (which can be seen in the eye-tracker as huge amplitude movements or as large eye-movements in degrees of visual angle; see Author response image 3 below to show a blink in the MEG data and the according eye-tracker data in degrees of visual angle). We will clarify this in the methods section.

As for the concern closed eyes to ignore visual distractors, in both experiments we can observe highly significant distractor cost in accuracy for visual distractors, which we hope will convince the reviewer that our visual distractors were working as intended.

Author response image 3.

Illustration of eye-tracker data for a trial without and a trial with a blink. All data points recorded during this trial are plotted. A, ICA component 1, which reflects blinks and its according data trace in a trial. No blink is visible. B, eye-tracker data transformed into degrees of visual angle for the trial depicted in A. C, ICA component 1, which reflects blinks and its according data trace in a trial. A clear blink is visible. D, eye-tracker data transformed into degrees of visual angle for the trial depicted in C.



(6) It would be interesting to examine the neutral cue trials in this task. For example, comparing auditory vs visual vs neutral cue conditions would be indicative of whether alpha was actively recruited or actively suppressed. In addition, comparing spectral activity during cue-to-target period on neutral-cue auditory correct vs incorrect trials

should mimic the comparison of auditory-cue vs visual-cue trials. Likewise, neutral-cue visual correct vs incorrect trials should mimic the attention-related differences in visual-cue vs auditory-cue trials.

We thank the reviewer for this suggestion. We have analysed the neutral cue trials in the EEG dataset (see suppl. Fig. 1) and will expand this figure to show all conditions. There were no significant differences to auditory or visual cues, but descriptively alpha power was higher for neutral cues compared to visual cues and lower for neutral cues compared to auditory cues. While this may suggest that for visual trials alpha is actively suppressed and for auditory trials actively recruited, we do not feel comfortable to make this claim, as the neutral condition may not reflect a completely neutral state. The neutral task can still be difficult, especially because of the uncertainty of the target modality.

As for the analysis of incorrect versus correct trials, we love the idea, but unfortunately the accuracy rate was quite high so that the number of incorrect trials would not be sufficient to perform a reliable analysis.

(7) In the abstract, the authors state that "This implies that alpha modulation does not solely regulate 'gain control' in early sensory areas but rather orchestrates signal transmission to later stages of the processing stream." However, I don't see any supporting evidence for the latter claim, that alpha orchestrates signal transmission to later stages of the processing stream. If the authors are claiming an alternative function to alpha, this claim should be strongly substantiated.

We thank the reviewer for pointing out, that we have not sufficiently explained our case. The first point refers to gain control akin to the alpha inhibition hypothesis, which claims that increases in alpha disengage a whole cortical area. Since we have confirmed the alpha increase in our data to originate from primary visual cortex through source analysis, this should lead to decreased visual processing. The increase in 36 Hz visual processing therefore directly contradicts the alpha inhibition hypothesis. We propose an alternative explanation for the functionality of alpha activity in this task. Through pulsed inhibition, information packages of relevant visual information could be transmitted down the processing stream, thereby enhancing relevant visual signal transmission. We believe the fact that the enhanced visual 36 Hz signal we found correlated with visual alpha power on a trial-by-trial basis, and did not originate from primary visual cortex, but from areas known for sensory integration supports our claim.

We will make this point clearer in our revised manuscript.

Reviewer #2 (Public review):

Brickwedde et al. investigate the role of alpha oscillations in allocating intermodal attention. A first EEG study is followed up with a MEG study that largely replicates the pattern of results (with small to be expected differences). They conclude that a brief increase in the amplitude of auditory and visual stimulus-driven continuous (steady-state) brain responses prior to the presentation of an auditory - but not visual - target speaks to the modulating role of alpha that leads them to revise a prevalent model of gating-by-inhibition.

Overall, this is an interesting study on a timely question, conducted with methods and analysis that are state-of-the-art. I am particularly impressed by the author's decision to replicate the earlier EEG experiment in MEG following the reviewer's comments on the original submission. Evidently, great care was taken to accommodate the reviewer's suggestions.

We thank the reviewer for the positive feedback and expression of interest in the topic of our manuscript.

Nevertheless, I am struggling with the report for two main reasons: It is difficult to follow the rationale of the study, due to structural issues with the narrative and missing information or justifications for design and analysis decisions, and I am not convinced that the evidence is strong, or even relevant enough for revising the mentioned alpha inhibition theory. Both points are detailed further below.

We thank the reviewer for raising this important point. We will revise our introduction and results in line with the reviewer's suggestions, hoping that our rationale will then be easier to follow and that our evidence will be more convincing.

Strength/relevance of evidence for model revision: The main argument rests on 1) a rather sustained alpha effect following the modality cue, 2) a rather transient effect on steady-state responses just before the expected presentation of a stimulus, and 3) a correlation between those two. Wouldn't the authors expect a sustained effect on sensory processing, as measured by steady-state amplitude irrespective of which of the scenarios described in Figure 1A (original vs revised alpha inhibition theory) applies? Also, doesn't this speak to the role of expectation effects due to consistent stimulus timing? An alternative explanation for the results may look like this: Modality-general increased steady-state responses prior to the expected audio stimulus onset are due to increased attention/vigilance. This effect may be exclusive (or more pronounced) in the attend-audio condition due to higher precision in temporal processing in the auditory sense or, vice versa, too smeared in time due to the inferior temporal resolution of visual processing for the attend-vision condition to be picked up consistently. As expectation effects will build up over the course of the experiment, i.e., while the participant is learning about the consistent stimulus timing, the correlation with alpha power may then be explained by a similar but potentially unrelated increase in alpha power over time.

We thank the reviewer for raising these insightful questions and suggestions.

It is true that our argument rests on a rather sustained alpha effect and a rather transient effect on steady-state responses and a correlation between the two. However, this connection would not be expected under the alpha inhibition hypothesis, which states that alpha activity would inhibit a whole cortical area (when irrelevant to the task), exerting "gain control". This notion directly contradicts our results of the "irrelevant" visual information a) being transmitted at all and b) increasing.

However, it has been shown on many occasions that alpha activity exerts pulsed inhibition, so we proposed an alternative theory of an involvement in signal transmission. In this case, the cyclic inhibition would serve as an ordering system, which only allows for high-priority information to pass, resulting in higher signal-to-noise. We do not make a claim about how fast or when these signals are transmitted in relation to alpha power. For instance, it could be that alpha power increases as a preparatory state even before signal is actually transmitted. Zhigalov (2020 Hum. Brain M.) has shown that in V1, frequency-tagging responses were up- and down regulated with attention – independent of alpha activity.

But we do believe that the fact that visual alpha power correlates on a trial-by-trial level with visual 36 Hz frequency-tagging increases and (a relationship which has not been found in V1, see Zhigalov 2020, Hum. Brain Mapp.) suggest a strong connection. Furthermore, the fact that the alpha modulation originates from early visual areas and occurs prior to any frequency-tagging changes, while the increase in frequency-tagging can be observed in areas which are

later in the processing stream (such as the precuneus) is strongly indicative for an involvement of alpha power in the transmission of this signal. We cannot fully exclude alternative accounts and mechanisms which effect both alpha power and frequency-tagging responses.

We do believe that the alternative account described by the reviewer does not contradict our theory, as we do believe that the alpha power modulation may reflect an expectation effect (and the idea that it could be related to the resolution of auditory versus visual processing is very interesting!). It is also possible that this expectation is, as the reviewer suggests, related to attention/vigilance and might result in a modality-general signal increase. And indeed, we can observe an increase in the frequency-tagging response in sensory integration areas. Accordingly, we believe that the alternative explanation provided by the reviewer contradicts the alpha inhibition hypothesis, but not necessarily our alternative theory.

We will revise the discussion, which we hope will make our case stronger and easier to follow. Additionally, we will mention the possibility for alternative explanations as well as the possibility, that alpha networks fulfil different roles in different locations/task environments.

Structural issues with the narrative and missing information: Here, I am mostly concerned with how this makes the research difficult to access for the reader. I list the major points below:

In the introduction the authors pit the original idea about alpha's role in gating against some recent contradictory results. If it's the aim of the study to provide evidence for either/or, predictions for the results from each perspective are missing. Also, it remains unclear how this relates to the distinction between original vs revised alpha inhibition theory (Fig. 1A). Relatedly if this revision is an outcome rather than a postulation for this study, it shouldn't be featured in the first figure.

We agree with the reviewer that we have not sufficiently clarified our goal as well as how different functionalities of alpha oscillations would lead to different outcomes. We will revise the introduction and restructure the results and hope that it will be easier to follow.

The analysis of the intermodulation frequency makes a surprise entrance at the end of the Results section without an introduction as to its relevance for the study. This is provided only in the discussion, but with reference to multisensory integration, whereas the main focus of the study is focussed attention on one sense. (Relatedly, the reference to "theta oscillations" in this sections seems unclear without a reference to the overlapping frequency range, and potentially more explanation.) Overall, if there's no immediate relevance to this analysis, I would suggest removing it.

We thank the reviewer for pointing this out and will add information about this frequency to the introduction part. We believe that the intermodulation frequency analysis is important, as it potentially supports the notion that condition differences in the visual-frequency tagging response are related to downstream processing rather than overall visual information processing in V1. We would therefore prefer to leave this analysis in the manuscript.

Reviewer #3 (Public review):

Brickwedde et al. attempt to clarify the role of alpha in sensory gain modulation by exploring the relationship between attention-related changes in alpha and attention-related changes in sensory-evoked responses, which surprisingly few studies have examined given the prevalence of the alpha inhibition hypothesis. The authors use robust methods and provide novel evidence that alpha likely exhibits inhibitory control over

later processing, as opposed to early sensory processing, by providing source-localization data in a cross-modal attention task.

This paper seems very strong, particularly given that the follow-up MEG study both (a) clarifies the task design and separates the effect of distractor stimuli into other experimental blocks, and (b) provides source-localization data to more concretely address whether alpha inhibition is occurring at or after the level of sensory processing, and (c) replicates most of the EEG study's key findings.

We are very grateful to the reviewer for their positive feedback and evaluation of our work.

There are some points that would be helpful to address to bolster the paper. First, the introduction would benefit from a somewhat deeper review of the literature, not just reviewing when the effects of alpha seem to occur, but also addressing how the effect can change depending on task and stimulus design (see review by Morrow, Elias & Samaha (2023)).

We thank the reviewer for this suggestion and agree. We will add a paragraph to the introduction which refers to missing correlation studies and the impact of task design.

Additionally, the discussion could benefit from more cautionary language around the revision of the alpha inhibition account. For example, it would be helpful to address some of the possible discrepancies between alpha and SSEP measures in terms of temporal specificity, SNR, etc. (see Peylo, Hilla, & Sauseng, 2021). The authors do a good job speculating as to why they found differing results from previous cross-modal attention studies, but I'm also curious whether the authors think that alpha inhibition/modulation of sensory signals would have been different had the distractors been within the same modality or whether the cues indicated target location, rather than just modality, as has been the case in so much prior work?

We thank the reviewer for suggesting these interesting discussion points and will include a paragraph in our discussion which goes deeper into these topics.

Overall, the analyses and discussion are quite comprehensive, and I believe this paper to be an excellent contribution to the alpha-inhibition literature.

<https://doi.org/10.7554/eLife.106050.1.sa0>